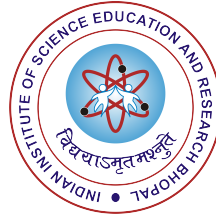


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Quantum Dense Coding Using Bipartite and Multipartite Entangled States Over Noiseless and Noisy Quantum Channels:

An IBM Quantum Experience

Project Report

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DECLARATION

Hereby I declare,that this paper is my original authorial work, which I have worked out by my own. All sources, references and literature used or excerpted during elaboration of this work are properly cited and listed in complete reference to the due source.

Rangan Chakrabarty
Kolkata
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ABSTRACT

Quantum information science has emerged as one of the most promising interdisciplinary research fields of the modern era. Among its various branches, quantum communication has attracted significant attention due to its potential to revolutionize secure and efficient information exchange. Quantum communication protocols that rely on pre-shared entangled states can generally be divided into two broad categories according to their objectives- classical information transfer and quantum state transfer.

Although quantum state transfer protocols have received significant attention, comparable efforts toward extending the practical applications of classical information transmission protocols using different multipartite entangled channels have received relatively limited attention in recent years. Most existing studies focus on proposing new entangled resources or evaluating protocol performance under ideal conditions, whereas comparatively fewer investigations address their practical implementation under realistic noisy environments. Furthermore, systematic comparisons among different entangled channels remain scarce.

Therefore, in this work, a comprehensive comparative study of quantum dense coding protocols based on different entangled states is presented under both ideal and realistic noisy quantum environments. The protocols are first analyzed theoretically to evaluate various performance metrics. Their robustness is then investigated under realistic Pauli noise models. Finally, a systematic comparison is carried out to identify the strengths and limitations of different entangled channels using Qiskit. The results demonstrate that the analytical derivations are in exact agreement with the simulation results. Furthermore, different protocols exhibit varying levels of robustness under different Pauli noise models, while increasing circuit depth and gate complexity make the protocols more error-prone. Altogether, the presented results establish a comprehensive framework for assessing quantum dense coding protocols, thereby facilitating their practical implementation in future quantum communication networks.

KEYWORDS

Quantum dense coding, quantum communication, Bell state, GHZ state, cluster States, noisy environment, bit flip channel, phase flip channel, bit-phase flip channel, dense coding capacity, state fidelity, QBER

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Chapter 1

Introduction

The wake of quantum information science [1, 2] over the past few decades is substantially increased due to improvement of different quantum communication technologies [3, 4, 5, 6, 7] where choosing wide varieties of entangled states [8, 9] play a significant role. The quantum communication protocols relying on a pre-shared entangled state can generally be divided into two broad categories, according to their objectives - classical information transfer [7, 10, 5, 11] and quantum state transfer [6]. Quantum entanglement plays an important role in formulation of various breakthrough technologies such as quantum dense coding [10] and quantum teleportation [6]. In the conventional quantum dense coding protocol [10], it is seen that the deterministic transmission of information can be achieved only when a maximally entangled state was pre-shared between communicating parties. Later on, quantum state transfer protocols were extended to different multipartite entangled channels [12, 13, 14, 15, 16, 17, 18] and subsequently to the framework of the quantum internet, where the reliable transfer of quantum information between distant nodes became a primary objective. Various approaches were proposed to realize these schemes in bidirectional and multidirectional communication framework, leading to the emergence of the rapidly growing field of quantum networks [19, 20]. Considerable research has since been devoted to improve the scalability, resistance against eavesdropping, and fault tolerance of these quantum communication protocols. In particular, the development of efficient entanglement distribution strategies, quantum repeaters, and network routing protocols has accelerated progress toward large-scale quantum communication infrastructures.

Amongst the various quantum communication procedures, like Quantum Teleportation (QT), Quantum Information Splitting (QIS), Quantum Dense Coding (QDC), Quantum Key Distribution (QKD), and Quantum Secure Direct Communication (QSDC), *QDC* emerged as one of the most significant demonstrations of the advantages offered by quantum entanglement. Since its original proposal, dense coding has been extensively generalized to multipartite entangled systems and higher-dimensional quantum systems. These multipartite entangled channels offer increased communication capacity, improved network connectivity, and greater flexibility for implementing distributed quantum communication protocols [21, 22, 23, 24, 25]. In particular, cluster states have attracted considerable attention because of their rich entanglement structure and their applications in measurement-based quantum computation and quantum communication networks [26, 27].

Despite these theoretical advancements, practical implementations of quantum dense coding remain challenging due to unavoidable harsh environmental conditions. Such interactions introduce rapid decoherence and quantum noise, which degrade entanglement and consequently reduce the reliability and efficiency of quantum communication protocols. *Entanglement Sudden Death (ESD)* phenomenon also deserves great attention in this regard. Among the most common noise mechanisms encountered in realistic quantum devices are primarily the *Pauli Errors*, including bit-flip, phase-flip, and bit-phase-flip errors, each of which alters the transmitted quantum information in different ways. Understanding the influence of these noise channels on dense coding protocols is therefore essential for developing robust quantum communication systems [1, 28].

The rapid development of superconducting quantum processors has significantly accelerated experimental research in quantum information processing. Cloud-accessible quantum computers provided by IBM Quantum, together with the Quantum Information Software Kit (Qiskit) software development framework (SDK), now enable researchers to simulate remotely and to implement a wide range of quantum communica-

tion protocols with decomposed gate structure on real quantum hardware. These platforms provide valuable opportunities to investigate the effects of hardware imperfections, gate errors, and environmental noise on practical quantum communication schemes [29].

The present work investigates the implementation and performance of quantum dense coding protocols using several entangled quantum channels, namely the Bell state, the three-qubit GHZ state, and four-, five-, and six-qubit cluster states. They are first analyzed under noiseless conditions to verify deterministic information transmission. Subsequently, their performance is examined under three important quantum noise models told previously. Numerical simulations are done using the QASM simulator of the Qiskit framework (v2.4.2), with selected implementations designed to be compatible with IBM superconducting quantum processors.

To provide a comprehensive comparison of the different entangled channels, several quantitative performance metrics are evaluated throughout this work. These include the dense coding capacity, Schmidt decomposition, Holevo bound, output state fidelity, channel efficiency, quantum bit error rate, and quantum resource cost. Moreover, the variation of state fidelity against the noise parameter is also studied in this work. The comparative analysis highlights the communication capabilities, robustness, and implementation complexity of each protocol, providing useful insights into the suitability of different multipartite entangled states for practical quantum dense-coding applications.

The remainder of this report is organized as follows. Chapter 2 presents the theoretical and mathematical background, including quantum entanglement, entangled quantum channels, density matrix formalism, dense coding performance metrics, and quantum noise models. Chapter 3 discusses the implementation of quantum dense coding protocols under noiseless conditions. Chapter 4 extends these protocols to noisy quantum channels by incorporating bit-flip, phase-flip, and bit-phase-flip noise models. Chapter 5 presents the results, a comparative analysis, and concluding discussions. Finally, the report concludes by highlighting the major findings and some potential future scopes & further research opportunities in the field of quantum secure communication.

Chapter 2

Background & Preliminaries

This first chapter of the report highlights the prerequisites and foundations required for understanding the quantum dense coding protocols. It begins with a brief discussion on quantum entanglement and its classification, bipartite and multipartite quantum entanglement, followed by a discussion of the entangled quantum channels used here, including Bell, GHZ, and multiqubit cluster states. The density matrix formalism is then introduced to describe quantum systems and their evolution under noisy environments. Subsequently, several important dense coding performance metrics are discussed. Finally, different pauli error models are also discussed on a short note. These concepts form the theoretical understanding to realise how quantum dense coding protocol works, how a message shared between two parties using minimal requirements of resources, what are the challenges faced during its implementation in real hardware system, till what extent such a protocol for different entangled channels is robust, and how such a noisy protocol can be rectified. Some of these questions are addressed in the subsequent sections of this report while the remaining questions are discussed by providing possible future research directions.

2.1 Entanglement and it's classification

In this section, the definition for pure entangled states will be introduced and afterwards it will be classified into different types. Since entangled states are usually defined as the complement of separable states, the definition of separable pure states and product states will precede the definition of entangled pure states.

Definition 1. Let $\mathcal{H}^A, \mathcal{H}^B$ be the Hilbert Spaces and let $|\phi\rangle \in \mathcal{H}^A \otimes \mathcal{H}^B$ be a pure state. The state $|\phi\rangle$ is a product state in $\mathcal{H} = \mathcal{H}^A \otimes \mathcal{H}^B$ if there are pure states such that $|\phi\rangle^A \in \mathcal{H}^A, |\phi\rangle^B \in \mathcal{H}^B$ such that

$$|\phi\rangle = |\phi\rangle^A \otimes |\phi\rangle^B \quad (2.1)$$

Thus a pure state is separable only if it is a product state in $\mathcal{H}$. Other the state is said to be entangled. It can be classified broadly into two categories- Bipartite entanglement and Multipartite entanglement. In case of bipartite entanglement the quantum entanglement shared between two distinct quantum subsystems; whereas in case of multipartite entanglement three or more quantum subsystems acting at the same time.

2.2 Entangled quantum channels

Entangled quantum channels are shared entangled quantum states distributed between communicating parties, typically Alice and Bob, and serve as the fundamental resource for quantum dense coding protocols. Depending on the structure of the shared entangled state, different types of quantum channels can be formulated, such as Bell state channels, GHZ state channels, W state channels, cluster state channels etc. The inherent entanglement properties of these channels determine important characteristics and performance matrices of the protocol.

2.2.1 Bell state quantum channel

The Bell state quantum channel is the simplest and most widely used entangled quantum channel for quantum dense coding protocol. The channel consists of two maximally entangled qubits that are shared between two distant parties, sender Alice and receiver Bob.

The four Bell states, each maximally entangled, form a complete orthonormal basis for the two-qubit Hilbert space and are given by

$$|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (2.2)$$

$$|\phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \quad (2.3)$$

$$|\psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad (2.4)$$

$$|\psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \quad (2.5)$$

According to Bennet & Wiesner [3] the bell state in eq(2.2) is commonly chosen as the transmission channel for Bell states dense coding.

Although using Bell states is the simplest way to realize quantum dense coding, multipartite entangled states can further extend the communication protocol having more communication capacities. The next subsection therefore introduces the GHZ-state quantum channel, which generalizes bipartite entanglement to three or more qubits.

2.2.2 GHZ state quantum channel

GHZ state is an important class of multipartite entangled states and can be thought as an natural extension of the two qubit Bell channel to systems containing three or more qubits. GHZ states exhibit genuine multipartite entanglement involving all constituent qubits simultaneously. The 3 qubit GHZ channel used here is defined as

$$|GHZ_3\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) \quad (2.6)$$

The three qubits are distributed among the communicating parties before the communication process begins. Removal or measurement of any one qubit completely destroys the multipartite entanglement of GHZ state, unlike Bell state. Therefore, GHZ states are highly sensitive to decoherence and abrupt faulty measurements but enable deterministic dense coding protocols with larger encoding spaces compared to bipartite channels. There are other types of GHZ channels which can also be used in dense coding protocol; like the 4 qubit GHZ channel (eq(2.7)), non-maximally entangled GHZ channel (eq(2.8)), 3 qubit GHZ like channel (eq(2.9), eq(2.10)), GHZ class channel (eq(2.11)) etc.

$$|GHZ_4\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle) \quad (2.7)$$

$$|GHZ_{nonmax}\rangle = \alpha |000\rangle + \beta |111\rangle; |\alpha|^2 + |\beta|^2 = 1 \quad (2.8)$$

$$|GHZ_l\rangle = \frac{1}{2}(|001\rangle + |010\rangle + |100\rangle + |111\rangle) \quad (2.9)$$

$$|GHZ_l\rangle = \frac{1}{2}(|000\rangle + |011\rangle + |101\rangle + |110\rangle) \quad (2.10)$$

$$|GHZ_{cl}\rangle = \frac{1}{\sqrt{2}}(|0\phi 0\rangle + |1\phi' 1\rangle) \quad (2.11)$$

2.2.3 Cluster state quantum channel

Cluster states belongs to a special class of multipartite entangled state of the family of graph states. Although, entanglement of Bell and GHZ states is distributed globally, the cluster states possess a highly structured pattern of nearest-neighbor entanglement determined by an underlying graph. This unique entanglement concept makes cluster states the primary resource for measurement-based quantum computation.

The 4-(eq(2.12)), 5-(eq(2.13)), 6-qubit cluster channels(eq(2.14)) used here are defined as

$$|CS_4\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle) \quad (2.12)$$

$$|CS_5\rangle = \frac{1}{2}(|10000\rangle + |00111\rangle + |11100\rangle + |11011\rangle) \quad (2.13)$$

$$|CS_6\rangle = \frac{1}{2}(|000000\rangle + |000111\rangle + |111000\rangle + |111111\rangle) \quad (2.14)$$

Except these cluster states there are also different types of cluster states like 2 qubit cluster state, 3 qubit cluster states, DV cluster states, CV cluster states, SPT cluster states etc.

2.3 Density matrix formalism

Most of the practical quantum systems interact with their surrounding environment, introduces significant noises in the system, resulting in statistical mixtures of quantum states. In such situations, the state-vector formalism alone becomes incapable to provide a complete description of the system. The density matrix formalism therefore offers a more general framework sufficient for representing both pure and mixed quantum states. It plays a pivotal role in describing noisy quantum channels, quantum measurements, decoherence, and entanglement.

2.3.1 Density matrix

The density matrix (or density operator) ρ_{AB} provides a unified mathematical description of both pure and mix types of quantum states.

For a pure quantum state the density matrix is defined as

$$\rho = |\psi\rangle \langle\psi| \quad (2.15)$$

while for a mixed state consisting of pure states ($|\psi_i\rangle$) with certain probability p_i , the density matrix is given by

$$\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i| \quad (2.16)$$

with

$$\sum_i p_i = 1$$

2.3.2 Partial trace

The partial trace is a mathematical operation used to eliminate the degrees of freedom of one subsystem while retaining the state of the remaining subsystem. It provides the bridge between the density matrix of a composite quantum system and the reduced density matrix of its individual subsystems.

2.3.3 Reduced density matrix

In quantum communication, one is often interested in describing only a subsystem of a larger composite quantum system. For example, in a dense coding protocol, Alice and Bob each possess only a part of an entangled state. The state of an individual subsystem is obtained through the reduced density matrix.

If a bipartite density operator of the composite system is ρ_{AB} then the reduced density matrix of Alice's subsystem is obtained by tracing over Bob's degrees of freedom,

$$\rho_A = \text{Tr}_B(\rho_{AB}) \quad (2.17)$$

whereas, bob's reduced density matrix is given by

$$\rho_B = \text{Tr}_A(\rho_{AB}) \quad (2.18)$$

For maximally entangled Bell states, the reduced density matrix of either qubit becomes

$$\rho_A = \rho_B = \frac{I}{2}$$

indicating that each maximally entangled qubit is therefore maximally mixed, despite the original system itself is a pure entangled state. Reduced density matrices are widely used for calculating subsystem entropy, Schmidt coefficients, dense coding capacity, and various entanglement measures.

2.4 Performance metrics for quantum dense coding

Quantum dense coding protocols are evaluated using several quantitative measures that characterize their information transmission capability, communication efficiency, robustness and resource utilization. In this work, the dense coding protocols for different entangled channels are analyzed using the following metrics.

2.4.1 Dense coding capacity

Dense coding capacity denotes the maximum amount of classical information that can be transmitted per qubit transmission through an entangled quantum channel.

For a shared bipartite quantum state ρ_{AB} the dense coding capacity is given by

$$C = \log_2(d_A) + S(\rho_B) - S(\rho_{AB}) \quad (2.19)$$

where,

d_A is the dimension of Alice's subsystem

$S(\rho) = -\text{Tr}(\rho \log_2 \rho)$ is the Von-Neumann Entropy

ρ_B is the Bob's reduced density matrix

ρ_{AB} is the density matrix of the shared quantum channel

2.4.2 Schmidt Decomposition

The Schmidt decomposition is a mathematical tool used to characterize the entanglement of pure bipartite quantum states. Any bipartite state can be expressed in the form

$$|\psi\rangle = \sum_{i=1}^r \lambda_i |u_i\rangle_A \otimes |v_i\rangle_B \quad (2.20)$$

where,

λ_i are the schmidt coefficients

$|u_i\rangle_A$ and $|v_i\rangle_B$ form the orthonormal basis

r is the schmidt rank

2.4.3 Holevo bound

The Holevo bound determines the maximum amount of classical information that can be extracted from a quantum ensemble. It provides an upper limit for the accessible classical information that can be obtained through quantum measurements. It is defined as

$$\chi = S(\rho) - \sum_i p_i S(\rho_i) \quad (2.21)$$

where,

$$\rho = \sum_i p_i \rho_i$$

p_i is the probability of transmitting the quantum state ρ_i
 $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$ is the Von-Neumann Entropy

2.4.4 Output state fidelity

Output state fidelity quantifies the accuracy between the ideal decoded quantum state and the experimentally obtained state after transmission through a noisy quantum channel. It is a direct metric to measure robustness of quantum communication protocol.

For an ideal pure state $|\psi_{ideal}\rangle$ and experimentally obtained density matrix ρ_{out} the output fidelity (F) is

$$F = \langle \psi_{ideal} | \rho_{out} | \psi_{ideal} \rangle \quad (2.22)$$

2.4.5 Channel efficiency

Channel efficiency quantifies the amount of classical information actually transmitted per transmitted quantum particle. It provides a useful comparison between different dense coding protocols employing different entangled resources.

It is defined as

$$\zeta = \frac{b_t}{q_t} \quad (2.23)$$

where,

b_t is the number of classical bits of information actually transmitted

q_t is the number of qubits actually transmitted to carry that classical message

2.4.6 Robustness of the protocol

The robustness of the applied dense coding protocol is defined as the average fidelity over the entire range of noise parameter ($0 \leq \eta \leq 1$) i.e.

$$\bar{F} = \int_0^1 F(\eta) d\eta$$

2.4.7 Quantum circuit depth

the circuit depth is the length of the longest dependency path through the quantum circuit. It measures how many sequential layers of operations are required to execute the circuit, assuming gates acting on different qubits can be performed simultaneously.

Mathematically, circuit depth = number of sequential gate layers

2.4.8 Quantum Bit Error Rate (QBER)

The Quantum Bit Error Rate (QBER) represents the fraction of transmitted classical bits that are incorrectly decoded due to quantum noise or operational errors to the total number of decoded transmitted classical message. It is defined as

$$QBER = \frac{N_{error}}{N_{total}} \quad (2.24)$$

where,

N_{error} is the number of incorrectly decoded bits,

N_{total} is the number of transmitted bits.

QBER also ranges from 0 to 1, although a lower QBER is always preferred in any communication protocol.

2.4.9 Quantum resource cost

The quantum resource cost quantifies the physical resources required to implement a dense coding protocol. Quantum resource cost often useful for comparing quantum channels with different employed entanglement capabilities.

2.5 Noisy quantum channels

In practical quantum communication systems, transmitted quantum information cannot be ideally perfect as during transmission, quantum states inevitably interact with their surrounding environment, resulting in decoherence and operational errors that degrade the transmitted information. These unwanted interactions are collectively referred to as quantum noise.

The evolution of an open quantum system is often described by using the *kraus operator*. The evolution of density matrix (quantum sum representation) under the effect of kraus operator is defined as

$$\phi(\rho) = \sum_{k=1}^n A_k \rho A_k^\dagger \quad (2.25)$$

If these operation elements satisfies the completeness relation $\sum_{k=1}^n A_k A_k^\dagger = I$ then it is called *trace preserving*; else if it does not satisfies the completeness relation i.e. $\sum_{k=1}^n A_k A_k^\dagger < I$ then it is called *non-trace preserving*. In case of trace preserving, ρ is often called density matrix of the system.

Here,

$$A_k = \langle e_k | u | e_0 \rangle \quad (2.26)$$

is called the kraus operator which acts on the physical system; where $|e_0\rangle$ is called the basis of environment and $|e_k\rangle$ is called the evolved state of environment.

Therefore, the action of a particular noise model on particular qubit k is defined as

$$\phi(\rho_k)_{noisy} = \sum_{j=1}^n A_j \rho_k A_j^\dagger \quad (2.27)$$

There are different types of noises which can interfere during transmission of qubits from one party to another, such as pauli noise channels, non-pauli or thermal noise, measurement or hardware noise etc. Here, primarily pauli noise channels and it's different types are discussed and implemented.

2.5.1 Bit flip channel

In this model, the computational basis states are interchanged with probability η_B , while the qubit remains unchanged with probability $(1 - \eta_B)$.

The evolution of density matrix is defined as

$$\phi(\rho) = (1 - \eta_B)\rho + \eta_B X \rho X \quad (2.28)$$

Here the kraus operators (ref. eq(2.26))are

$$A_0 = \sqrt{\eta_B} I \quad (2.29)$$

and,

$$A_1 = \sqrt{1 - \eta_B} X \quad (2.30)$$

where, $0 \leq \eta_B \leq 1$ and

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is the pauli X operator.

2.5.2 Phase flip channel

The phase-flip channel models the errors that preserve the computational basis states while reversing the relative phase difference between them.

If probability of occurring phase flip is η_W and that of remaining it unchanged is $(1 - \eta_W)$ then the evolution of density matrix is defined as

$$\phi(\rho) = (1 - \eta_W)\rho + \eta_W Z\rho Z \quad (2.31)$$

Here the kraus operators (ref. eq(2.26))are

$$A_0 = \sqrt{\eta_W} I \quad (2.32)$$

and,

$$A_1 = \sqrt{1 - \eta_W} Z \quad (2.33)$$

where, $0 \leq \eta_W \leq 1$ and

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the pauli Z operator.

2.5.3 Bit-phase flip channel

The bit-phase-flip channel combines the effects of both bit-flip and phase-flip errors. If probability of occurring bit-phase flip is η_F and that of remaining it unchanged is $(1 - \eta_F)$ then the evolution of density matrix is defined as

$$\phi(\rho) = (1 - \eta_F)\rho + \eta_F Y\rho Y \quad (2.34)$$

Here the kraus operators (ref. eq(2.26))are

$$A_0 = \sqrt{\eta_F} I \quad (2.35)$$

and,

$$A_1 = \sqrt{1 - \eta_F} Y \quad (2.36)$$

where, $0 \leq \eta_F \leq 1$ and

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

is the pauli Y operator.

With the necessary theoretical foundations established, the focus now shifts towards the next chapter which presents the implementation of quantum dense coding protocols employing Bell, GHZ, and multiqubit cluster-state quantum channels under ideal noiseless conditions. The encoding and decoding strategies for each protocol are developed systematically and discussed wherever necessary, providing a framework for the subsequent investigation of their behavior under various quantum noise models.

Chapter 3

Quantum dense coding protocols in noiseless environment

This chapter presents the formulation, implementation and analysis of quantum dense coding protocols using different entangled channels under ideal noiseless conditions. The absence of environmental interactions and decoherence allows the intrinsic communication capability of each entangled quantum channel to be investigated without the influence of quantum noise. Establishing such an ideal framework is essential, as it provides the theoretical benchmark against which the performance of noisy protocols can later be compared. The correctness of each protocol is verified through quantum circuit simulations using the Qiskit software development framework.

The chapter begins with the conventional Bell-state dense coding protocol and subsequently extends the implementation to multipartite entangled channels, including the three-qubit GHZ state and the four-, five-, and six-qubit cluster states.

3.1 Standard QDC using Bell state

The Bell-state quantum channel provides the simplest realization of the quantum dense coding protocol and was first proposed by Bennett and Wiesner in 1992 [3]. This demonstrates one of the most significant advantages of quantum communication over its classical counterpart.

3.1.1 Preparation of Bell state

Initially, two qubits are considered to prepare the Bell state. The state can be represented as $|\psi_0\rangle = |0\rangle_A \otimes |0\rangle_B$. To prepare the Bell state, Alice applies two quantum gates respectively- Hadamard (H) gate, followed by Controlled-NOT (CNOT) gate.

$$|\psi_1\rangle = CNOT(A, B)(I_B \otimes H_A)(|0\rangle_A \otimes |0\rangle_B) \quad (3.1)$$

Application of these gates transforms the quantum state to the following state

$$|\psi_1\rangle = |\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (3.2)$$

This is the standard Bell state used for dense coding mechanism. Now, the first qubit is kept by Alice and the counterpart is sent to Bob; so that there is 1:1 distribution of qubits.

3.1.2 Dense coding Capacity

Here, the dimension of Alice's system (d_A)=2; Reduced Von-neumann entropy =1, and Von-neumann entropy of pure state=0.

Therefore, from eq(2.19), DCC=2 bits and number of orthogonal codewords are $N^2 = 4$ i.e I,X,Y,Z. Also, the maximum possible schmidt rank of the system is 2.

3.1.3 Encoding operation of Alice

Now, Alice wants to send two bits of classical message (e.g. 00, 01, 10, 11) to Bob. For this operation she needs $2^2 = 4$ different single qubit unitary operations. These single qubit unitary operations are generally done using local single qubit pauli operators $U \in \{I, X, Y, Z\}$. Each of the encoding operators transforms the Bell state to a different encoded state. If she want to send message 00 then $U_{00}=(I \otimes I)$ operation is done

$$|\psi_2\rangle = (I \otimes I) |\psi_1\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\phi^+\rangle \quad (3.3)$$

Similarly if she wants to send message 01, 10 or 11 then respectively $U_{01}=(X \otimes I)$, $U_{10}=(Y \otimes I)$, $U_{11}=(Z \otimes I)$ operations are done.

$$|\psi_2\rangle = (X \otimes I) |\psi_1\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} = |\psi^+\rangle$$

$$|\psi_2\rangle = (Y \otimes I) |\psi_1\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |\psi^-\rangle$$

$$|\psi_2\rangle = (Z \otimes I) |\psi_1\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\phi^-\rangle$$

Each of these encoded state are mutually orthogonal. It ensures that, Bob can perfectly distinguish these states while decoding with certainty. After encoding the classical information, Alice sends the qubit in her possession to Bob; so that Bob now owns both of the qubits to perform decoding operations and measure them.

3.1.4 Bob's decoding operation

Bob now performs the decoding process by imposing the inverse of the Bell state preparation operation, consisting of a CNOT gate followed by a Hadamard gate. This decoding operation converts the Bell basis into the computational basis, allowing the encoded message to be recovered by a projective measurement. The decoding operation can be shown as

$$|\psi_3\rangle = (I_B \otimes H_A) CNOT(A, B) |\psi_2\rangle \quad (3.4)$$

If she sent message 00 then the decoded state will be

$$|\psi_3\rangle = (I_B \otimes H_A) CNOT(A, B) |\phi^+\rangle = |00\rangle \quad (3.5)$$

Similarly, if the sent message is 01, 10 or 11 then the final state will be respectively

$$|\psi_3\rangle = (I_B \otimes H_A) CNOT(A, B) |\psi^+\rangle = |01\rangle$$

$$|\psi_3\rangle = (I_B \otimes H_A) CNOT(A, B) |\psi^-\rangle = |10\rangle$$

$$|\psi_3\rangle = (I_B \otimes H_A) CNOT(A, B) |\phi^-\rangle = |11\rangle$$

As the messages are decoded with full accuracy, then the protocol can be called *deterministic*.

3.1.5 Quantum circuit of the protocol

The quantum circuit consisting of the bell state preparation, encoding operations, decoding operations and measurement is shown below

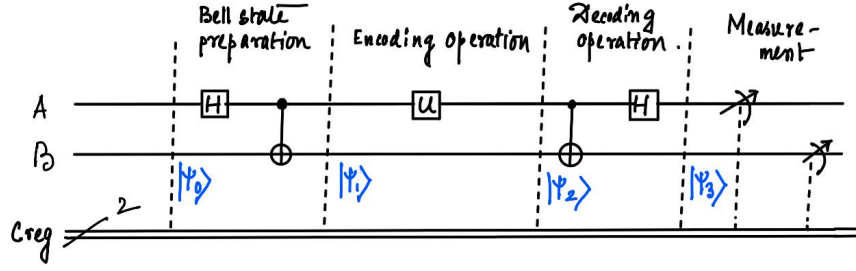


Figure 3.1: Circuit for QDC using 2 qubit Bell state

3.1.6 Implementation using Qiskit

The given circuit is implemented using QASM Simulator of IBM's Qiskit software for all the four possibilities and it is seen that for all the four cases the sent message can be decoded with maximal success probability. Here the corresponding circuit diagram and output result for the case 1 where classical message 00 is sent is shown.

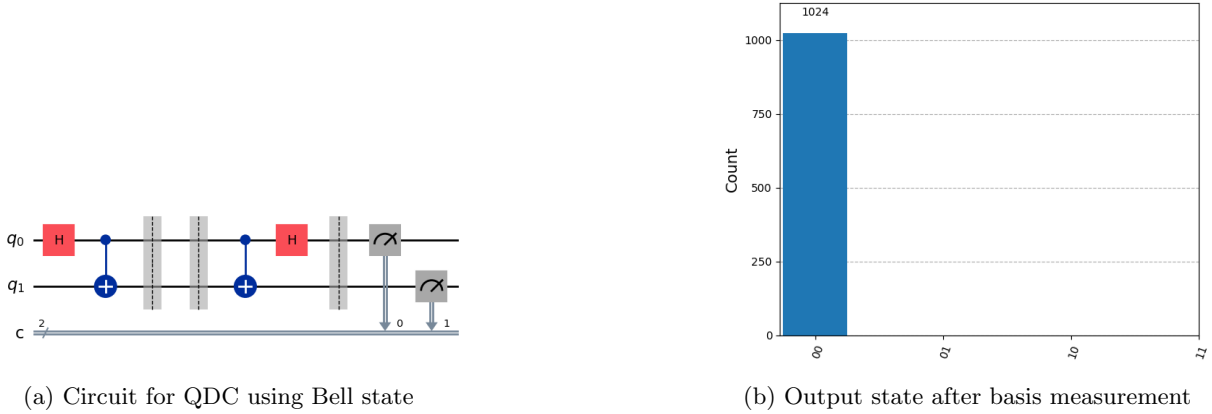


Figure 3.2: Implementation of QDC using Bell state Protocol when the message 00 is sent

3.2 QDC using 3 qubit GHZ state

The Greenberger-Horne-Zeilinger (GHZ) state provides one of the simplest realizations of multipartite quantum entanglement and serves as a natural extension of the Bell-state dense coding protocol. Unlike the Bell channel, the GHZ channel consists of three maximally entangled qubits, allowing quantum correlations to be shared among three subsystems simultaneously. The three-qubit GHZ-state quantum channel provides a larger encoding space than the Bell-state channel due to its multipartite entanglement structure. In the following subsections, the preparation of the GHZ state, the encoding and decoding procedures, the corresponding quantum circuit implementation, and the simulation results are discussed in detail.

3.2.1 Preparation of 3 qubit GHZ state

Firstly three qubits, all initialized to zero state are considered. The state represented as $|\psi_0\rangle = |0\rangle \otimes |0\rangle \otimes |0\rangle$. To prepare the GHZ state, Alice applies one Hadamard gate and two CNOT gates consecutively.

$$|\psi_1\rangle = CNOT(1, 3)CNOT(1, 2)(H_1 \otimes I_2 \otimes I_3)(|000\rangle) \quad (3.6)$$

Imposing these gates transforms the state into

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

This is the standard GHZ state used for the dense coding protocol here. After state preparation, first two qubit were kept by Alice herself and the 3rd one is transferred to Bob via the quantum channel.

3.2.2 Dense coding Capacity

Here, the dimension of Alice's system (d_A)=4; Reduced Von-neumann entropy =1, and Von-neumann entropy of pure state=0.

Therefore, from eq(2.19), DCC=3 bits and number of orthogonal codewords are $N^2 = 8$.

Also, the maximum possible schmidt rank of the system is 2 i.e. maximally entangled.

3.2.3 Encoding operations of Alice

Through this protocol Alice can send three qubits of message (e.g. 000, 001, 010, 011, 100, 101, 110, 111) to Bob. Therefore total 8 unitary operations are required which can be formulated as $U \in U_1(I, X, Y, Z) \otimes U_2(I, X)$. Alongside this scheme other schemes can also be used to encode messages where domain of single qubit encoding operators are different[2]. Total 8 unitary operations of this scheme and the resultant encoding state are shown here in the following table. Each of the encoded state are mutually orthogonal.

Table 3.1: Summary of encoding operations performed on 3 qubit GHZ state by Alice

S.No.	Encoding Unitary Operator	Encoded state ($ \psi_2\rangle$)	Classical encoded message
1.	$U_0 = I \otimes I$	$\frac{1}{\sqrt{2}}(000\rangle + 111\rangle)$	000
2.	$U_1 = X \otimes I$	$\frac{1}{\sqrt{2}}(100\rangle + 011\rangle)$	011
3.	$U_2 = Y \otimes I$	$\frac{1}{\sqrt{2}}(011\rangle - 100\rangle)$	111
4.	$U_3 = Z \otimes I$	$\frac{1}{\sqrt{2}}(000\rangle - 111\rangle)$	100
5.	$U_4 = I \otimes X$	$\frac{1}{\sqrt{2}}(010\rangle + 101\rangle)$	010
6.	$U_5 = X \otimes X$	$\frac{1}{\sqrt{2}}(110\rangle + 001\rangle)$	001
7.	$U_6 = Y \otimes X$	$\frac{1}{\sqrt{2}}(001\rangle - 110\rangle)$	101
8.	$U_7 = Z \otimes X$	$\frac{1}{\sqrt{2}}(010\rangle - 101\rangle)$	110

After selective encoding operation is done Alice sends the two qubits in her possession to Bob, so that now Bob possess all the 3 entangled qubits.

3.2.4 Decoding operations of Bob

Bob now performs the decoding process by consecutively applying inverse of GHZ state preparation operation. The final state obtained as

$$|\psi_3\rangle = (H_1 \otimes I_2 \otimes I_3)CNOT(1, 2)CNOT(1, 3)|\psi_2\rangle \quad (3.7)$$

Using this operation, for each encoded message, the exact classical encoded state as shown in table (3.1) is decoded successfully by Bob.

3.2.5 Quantum circuit of the protocol

The quantum circuit consisting of the GHZ state preparation, encoding operations, decoding operations and measurement is shown below

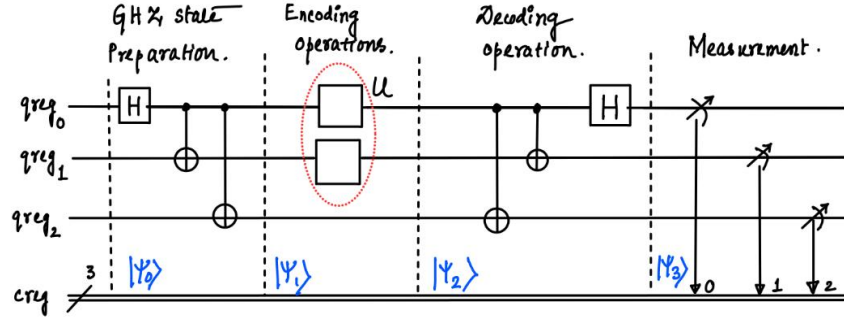


Figure 3.3: Circuit for QDC using 3 qubit GHZ state

3.2.6 Implementation using Qiskit

The given circuit is implemented using QASM Simulator of IBM's Qiskit software for all possibilities and it is seen that for all the cases the sent message can be decoded with maximal success probability. Here the corresponding circuit diagram and output result for the case 7 of the said scheme where classical message 101 is sent is shown.

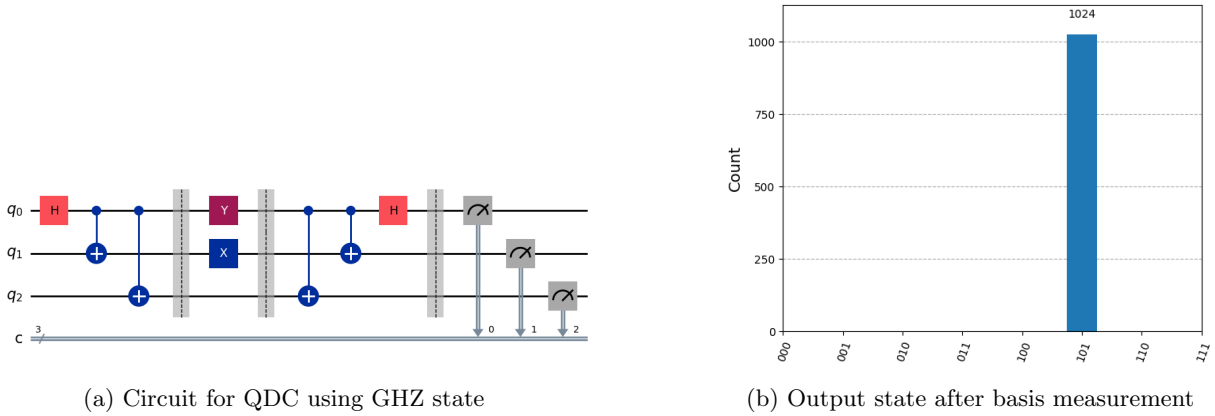


Figure 3.4: Implementation of QDC Protocol using GHZ state when the message 101 is sent

3.3 QDC using 4 qubit cluster state

Cluster states constitute a special class of multipartite entangled states belonging to the family of graph states. Unlike Bell and GHZ states, whose entanglement is shared globally among the participating qubits, cluster states exhibit a structured pattern of nearest-neighbour entanglement generated through controlled-phase interactions.

In this section, a four-qubit linear cluster state is employed as the entangled quantum channel for implementing the dense coding protocol. The protocol is discussed in detail below.

3.3.1 Preparation of 4 qubit cluster state

First and foremost, 4 qubits initialized to state $|\psi_0\rangle = |0000\rangle$ is considered. To prepare the desired 4 qubit cluster state, Alice applies a total of 6 Hadamard (H) gates and 3 Controlled-Z (CZ) gate. The gate operation will be like this

$$|\psi_1\rangle = (H_1 \otimes I \otimes I \otimes H_2)CZ_{3,4}CZ_{2,3}CZ_{1,2}H^{\otimes 4}|\psi_0\rangle \quad (3.8)$$

The initial state therefore transforms to

$$|\psi_1\rangle = |CS_4\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$$

When the state preparation is done, Alice keeps qubit 1 and 2 in her possession and sends the rest qubits to Bob.

3.3.2 Dense coding Capacity

Here, the dimension of Alice's system (d_A)=4; Reduced Von-neumann entropy =2, and Von-neumann entropy of pure state=0.

Therefore, from eq(2.19), DCC=4 bits and number of orthogonal codewords are $N^2 = 16$.

Also, the maximum possible schmidt rank of the system is 4.

3.3.3 Encoding operations of Alice

For each qubit in Alice's possession, she can apply four single qubit unitary operations $U \in \{I, X, Y, Z\}$. So, total $4^2 = 16$ encoding operations are possible to encode a message. All the possibilities are shown below.

But it is seen that all encoded states are *not* orthogonal with rest of the states. It can be proved using

Table 3.2: Summary of encoding operations performed on 4 qubit cluster state by Alice

S.No.	Encoding Unitary Operator	Encoded state ($ \psi_2\rangle$)	Classical encoded message
1.	$U_0 = I \otimes I$	$\frac{1}{2}(0000\rangle + 0011\rangle + 1100\rangle + 1111\rangle)$	0000
2.	$U_1 = I \otimes X$	$\frac{1}{2}(0100\rangle + 0111\rangle + 1000\rangle + 1011\rangle)$	1010
3.	$U_2 = I \otimes Y$	$\frac{1}{2}(- 0100\rangle - 0111\rangle + 1000\rangle + 1011\rangle)$	1110
4.	$U_3 = I \otimes Z$	$\frac{1}{2}(0000\rangle + 0011\rangle - 1100\rangle - 1111\rangle)$	0100
5.	$U_4 = X \otimes I$	$\frac{1}{2}(1000\rangle + 1011\rangle + 0100\rangle + 0111\rangle)$	1000
6.	$U_5 = X \otimes X$	$\frac{1}{2}(1100\rangle + 1111\rangle + 0000\rangle + 0011\rangle)$	0010
7.	$U_6 = X \otimes Y$	$\frac{1}{2}(- 1100\rangle - 1111\rangle + 0000\rangle + 0011\rangle)$	0110
8.	$U_7 = X \otimes Z$	$\frac{1}{2}(1000\rangle + 1011\rangle - 0100\rangle - 0111\rangle)$	1100
9.	$U_8 = Y \otimes I$	$\frac{1}{2}(- 1000\rangle - 1011\rangle + 0100\rangle + 0111\rangle)$	1100
10.	$U_9 = Y \otimes X$	$\frac{1}{2}(- 1100\rangle - 1111\rangle + 0000\rangle + 0011\rangle)$	0110
11.	$U_{10} = Y \otimes Y$	$\frac{1}{2}(1100\rangle + 1111\rangle + 0000\rangle + 0011\rangle)$	0010
12.	$U_{11} = Y \otimes Z$	$\frac{1}{2}(- 1000\rangle - 1011\rangle - 0100\rangle - 0111\rangle)$	1000
13.	$U_{12} = Z \otimes I$	$\frac{1}{2}(0000\rangle + 0011\rangle - 1100\rangle - 1111\rangle)$	0100
14.	$U_{13} = Z \otimes X$	$\frac{1}{2}(0100\rangle + 0111\rangle - 1000\rangle - 1011\rangle)$	1110
15.	$U_{14} = Z \otimes Y$	$\frac{1}{2}(- 0100\rangle - 0111\rangle - 1000\rangle - 1011\rangle)$	1010
16.	$U_{15} = Z \otimes Z$	$\frac{1}{2}(0000\rangle + 0011\rangle + 1100\rangle + 1111\rangle)$	0000

appropriate computational techniques that the gram matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

is not diagonal, indicating that all the states are not orthogonal. Furthermore, it is also computed that the non-orthogonal pairs are

```
II  ZZ  (0.9999999999999999+0j)
IX  ZY  0.9999999999999999j
IY  ZX  -0.9999999999999999j
IZ  ZI  (0.9999999999999999+0j)
XI  YZ  0.9999999999999999j
XX  YY  (-0.9999999999999999+0j)
XY  YX  (0.9999999999999999+0j)
XZ  YI  0.9999999999999999j
```

So, these pairs can't coexist for encoding operation (as $\langle \psi_i | \psi_j \rangle \neq 0$). Therefore, two sets of encoding operations exist which are themselves mutually orthogonal

Set 1: II, IX, IY, IZ, XI, XX, XY, XZ

or, Set 2: ZZ, ZY, ZX, ZI, YZ, YY, YX, YI

Either encoding operations of Set 1 or Set 2 can be applied for perfect deterministic dense coding mechanism. Here set 1 operations are used to encode message. When the selective encoding operation is done by Alice, the qubits in her possession are sent to Bob, so that he can now decode the encoded message.

3.3.4 Decoding operations of Bob

Bob, now spontaneously applies the inverse of 4 qubit cluster state preparation operation to decode the message. The final state will be

$$|\psi_3\rangle = H^{\otimes 4} CZ_{1,2} CZ_{2,3} CZ_{3,4} (H_1 \otimes I \otimes I \otimes H_2) |\psi_2\rangle \quad (3.9)$$

The final obtained state is measured in computational basis to recover the classical message successfully deterministically.

3.3.5 Quantum circuit

The quantum circuit consisting of the 4 qubit Cluster state preparation, encoding operations, decoding operations and measurement is shown below

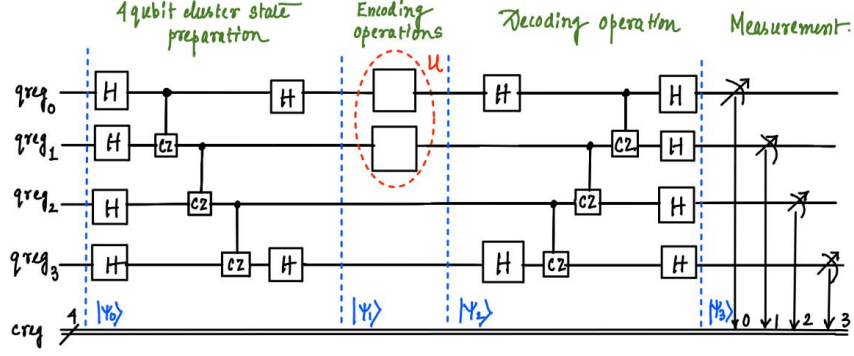
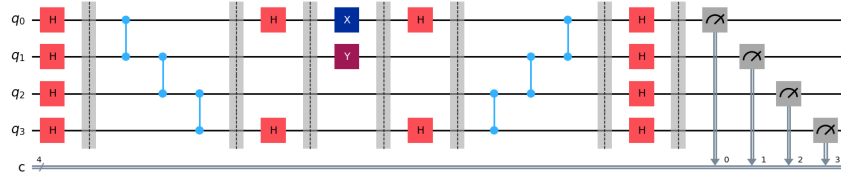


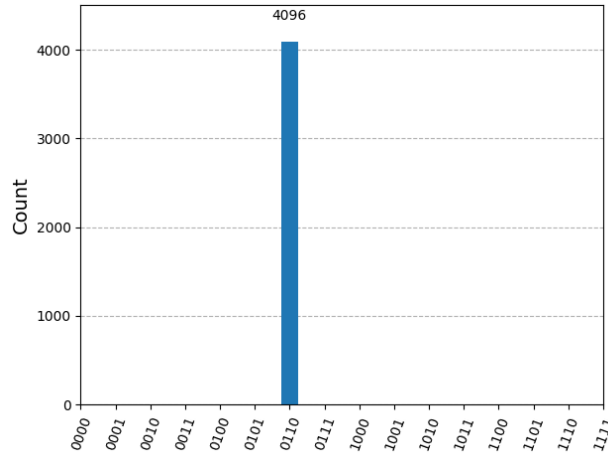
Figure 3.5: Circuit for QDC using 4 qubit Cluster state

3.3.6 Implementation using Qiskit

The given circuit is implemented using QASM Simulator of IBM's Qiskit software for all possibilities and it is seen that for all the eight cases the sent message can be decoded with maximal success probability. Here the corresponding circuit diagram and output result for the case 7 of the said scheme where classical message 0110 sent, is shown.



(a) Circuit for QDC using 4 qubit Cluster state



(b) Output state after basis measurement

Figure 3.6: Implementation of QDC Protocol using 4 qubit cluster state when the message 0110 is sent

3.4 QDC using 5 qubit cluster state

In the present section, a five-qubit linear cluster state is employed as the entangled quantum channel for implementing the quantum dense coding protocol. Compared with the Bell-state and three-qubit GHZ-

state protocols, the five-qubit cluster-state channel provides a significantly larger Hilbert space and a richer multipartite entanglement structure. Consequently, it supports a larger set of orthogonal encoding operations and offers greater flexibility for investigating the communication capability and robustness of quantum dense coding protocols. In the following subsections, the preparation of the five-qubit cluster state, the encoding and decoding procedures, the corresponding Qiskit implementation, and the simulation results are presented and discussed in detail.

3.4.1 Preparation of 5 qubit cluster state

Initially 5 qubits are considered to be initialized to state $|\psi_0\rangle = |00000\rangle$. To prepare the desired 5 qubit cluster state, Alice applies a total of 8 Hadamard (H) gates and 4 Controlled-Z (CZ) gate. The gate operation will be like this

$$|\psi_1\rangle = (H_1 \otimes I \otimes H_2 \otimes I \otimes H_3) CZ_{4,5} CZ_{3,4} CZ_{2,3} CZ_{1,2} H^{\otimes 5} |\psi_0\rangle \quad (3.10)$$

Theefore, the resultant state will be

$$|\psi_1\rangle = |CS_5\rangle = |00000\rangle + |11100\rangle + |11011\rangle + |00111\rangle$$

After the preparation of state Alice keeps first two qubit with her, while sends the rest qubits to Bob.

3.4.2 Dense coding Capacity

Here, the dimension of Alice's system (d_A)=4; Reduced Von-neumann entropy =2, and Von-neumann entropy of pure state=0.

Therefore, from eq(2.19), DCC=4 bits and number of orthogonal codewords are $N^2 = 16$.

Also, the maximum possible schmidt rank of the system is 4.

3.4.3 Encoding operations of Alice

For each qubit in Alice's possession, like the 4 qubit cluster state dense coding protocol in this case also, she can apply four single qubit unitary operations $U \in \{I, X, Y, Z\}$. So, total $4^2 = 16$ encoding operations are possible to encode a message. All the possibilities are shown below. Like in the previous case here also it is

Table 3.3: Summary of encoding operations performed on 5 qubit cluster state by Alice

S.No.	Encoding Unitary Operator	Encoded state ($ \psi_2\rangle$)	Classical encoded message
1.	$U_0 = I \otimes I$	$ 00000\rangle + 00111\rangle + 11100\rangle + 11011\rangle$	00000
2.	$U_1 = I \otimes X$	$ 01000\rangle + 10100\rangle + 01111\rangle + 10011\rangle$	10100
3.	$U_2 = I \otimes Y$	$- 01000\rangle + 10100\rangle - 01111\rangle + 10011\rangle$	11100
4.	$U_3 = I \otimes Z$	$ 00000\rangle + 00111\rangle - 11100\rangle - 11011\rangle$	01000
5.	$U_4 = X \otimes I$	$ 10000\rangle + 01011\rangle + 01100\rangle + 10111\rangle$	10000
6.	$U_5 = X \otimes X$	$ 11000\rangle + 11111\rangle + 00100\rangle + 00011\rangle$	00100
7.	$U_6 = X \otimes Y$	$- 11000\rangle + 00011\rangle - 11111\rangle + 00100\rangle$	01100
8.	$U_7 = X \otimes Z$	$ 10000\rangle - 01100\rangle - 01011\rangle + 10111\rangle$	11000
9.	$U_8 = Y \otimes I$	$- 10000\rangle + 01100\rangle + 01011\rangle - 10111\rangle$	11100
10.	$U_9 = Y \otimes X$	$- 11000\rangle + 00100\rangle + 00011\rangle - 11111\rangle$	01100
11.	$U_{10} = Y \otimes Y$	$ 11000\rangle + 00100\rangle + 00011\rangle + 11111\rangle$	00100
12.	$U_{11} = Y \otimes Z$	$- 10000\rangle - 01100\rangle - 01011\rangle - 10111\rangle$	10000
13.	$U_{12} = Z \otimes I$	$ 00000\rangle - 11100\rangle - 11011\rangle + 00111\rangle$	01000
14.	$U_{13} = Z \otimes X$	$ 01000\rangle - 10100\rangle - 10011\rangle - 01111\rangle$	11100
15.	$U_{14} = Z \otimes Y$	$- 01000\rangle - 10100\rangle - 10011\rangle - 01111\rangle$	10100
16.	$U_{15} = Z \otimes Z$	$ 00000\rangle + 11100\rangle + 11011\rangle + 00111\rangle$	00000

seen that all states are not orthogonal and it can be shown by the gram matrix

=====

GRAM MATRIX

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$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Encoded states are **not** mutually orthogonal.

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Similarly, using computational techniques the non-orthogonal pairs are also figured out.

=====

Non-Orthogonal Pairs

=====

II <-> ZZ
Overlap = (0.9999999999999987+0j)

IX <-> ZY
Overlap = 0.9999999999999987j

IY <-> ZX
Overlap = -0.9999999999999987j

IZ <-> ZI
Overlap = (0.9999999999999987+0j)

XI <-> YZ
Overlap = 0.9999999999999987j

XX <-> YY
Overlap = (-0.9999999999999987+0j)

XY <-> YX
Overlap = (0.9999999999999987+0j)

...

XZ <-> YI
Overlap = 0.9999999999999987j

Total Non-Orthogonal Pairs = 8

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Therefore, again there are two sets of encoding operator which are orthogonal amongst themselves only. Here, only the first set of encoding operators (II, IX, IY, IZ, XI, XX, XY and XZ) are considered for further calculations.

When encoding operation is completed, the first two qubits were sent to Bob via quantum channel.

3.4.4 Decoding operations of Bob

Bob, now spontaneously applies the inverse of 5 qubit cluster state preparation operation to decode the message. The final state will be

$$|\psi_3\rangle = H^{\otimes 5} CZ_{1,2} CZ_{2,3} CZ_{3,4} CZ_{4,5} (H_1 \otimes I \otimes H_2 \otimes I \otimes H_3) |\psi_2\rangle \quad (3.11)$$

3.4.5 Quantum Circuit

The quantum circuit consisting of the 5 qubit Cluster state preparation, encoding operations, decoding operations and measurement is shown below

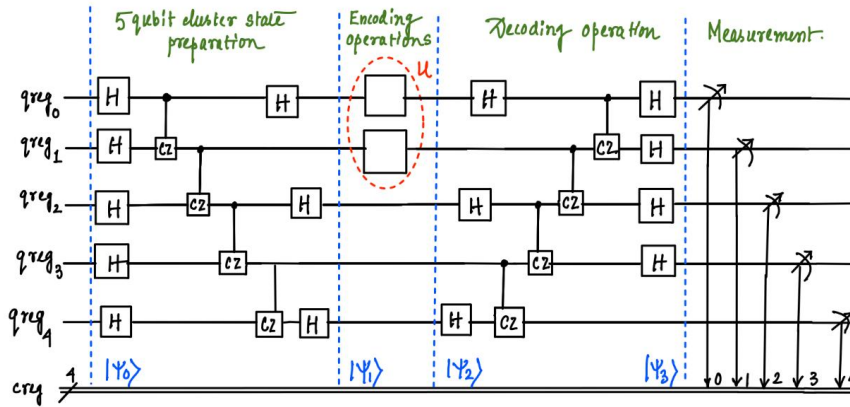
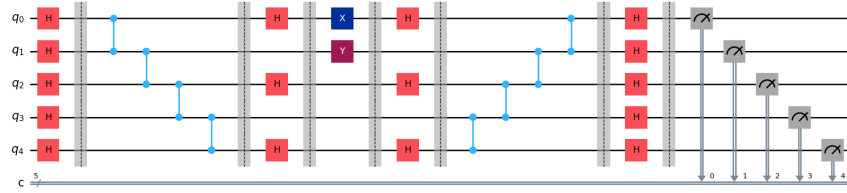


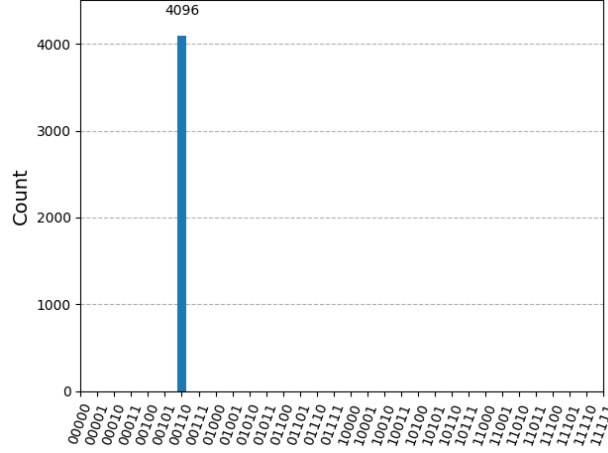
Figure 3.7: Circuit for QDC using 5 qubit Cluster state

3.4.6 Implementation using Qiskit

The drawn circuit is implemented using QASM Simulator of IBM's Qiskit software for all possibilities and it is seen that for all the eight cases the sent message can be decoded with maximal success probability. Here the corresponding circuit diagram and output result for the case 7 of the said scheme is shown.



(a) Circuit for QDC using 5 qubit cluster state



(b) Output state after basis measurement

Figure 3.8: Implementation of QDC Protocol using 5 qubit cluster state when the message 01100 is sent

3.5 QDC using 6 qubit cluster state

The six-qubit cluster state represents a highly entangled multipartite quantum channel that extends the principles of quantum dense coding beyond bipartite and small-scale multipartite systems. Owing to its rich entanglement structure and larger Hilbert-space dimension, the six-qubit cluster state provides a significantly larger encoding space, making it an attractive resource for high-capacity quantum communication protocols. Furthermore, cluster states possess a graph-based entanglement structure that has established them as fundamental resources for measurement-based quantum computation, quantum error correction, and various quantum communication tasks. Compared with the Bell-state, GHZ-state, and lower-dimensional cluster-state quantum channels, the six-qubit cluster state offers a substantially larger orthogonal encoding space and consequently supports higher information transmission capability. The increased number of entangled qubits also provides an ideal platform for investigating the scalability of quantum dense coding protocols and evaluating their robustness under various quantum noise models. In the following subsections, the preparation of the six-qubit cluster state, the encoding and decoding procedures, the corresponding Qiskit implementation, and the simulation results are discussed in detail.

3.5.1 Preparation of 6 qubit cluster state

Initially 6 qubits are considered to be initialized to state $|\psi_0\rangle = |000000\rangle$. To prepare the desired 6 qubit cluster state, Alice applies a total of 2 Hadamard (H) gates and 4 Controlled-NOT (CNOT) gates. The gate operation will be like this

$$|\psi_1\rangle = CNOT_{4,6}CNOT_{4,5}CNOT_{1,3}CNOT_{1,2}(H_1 \otimes I \otimes I \otimes H_2 \otimes I \otimes I) |\psi_0\rangle \quad (3.12)$$

Thefore, the resultant state will be

$$|\psi_1\rangle = |CS_6\rangle = \frac{1}{2}(|000000\rangle + |000111\rangle + |111000\rangle + |111111\rangle)$$

After the preparation of state Alice keeps the 1st, 6th and 4th qubit with her, while sends the rest qubits to Bob. This unique type of distribution of qubits will be justified by the discussion on dense coding capacity later.

3.5.2 Dense coding Capacity

Here, the dimension of Alice's system (d_A)=3.

Therefore, from eq(2.19), DCC ≈ 5 bits and number of orthogonal codewords are $N^2 = 32$.

Also, the maximum possible schmidt rank of the system is 8.

3.5.3 Encoding operations of Alice

For each qubit in Alice's possession, like the 4- and 5- qubit cluster state dense coding protocol in this case also, she can apply four single qubit unitary operations $U \in \{I, X, Y, Z\}$. So, total $4^3 = 64$ encoding operations are possible to encode a message.

But the dense coding capacity of the channel is $C=5$ (3.5.2) i.e. the largest possible subset of local pauli encoding operators should produce 32 mutually orthogonal encoding codewords in accordance with the DCC. Therefore maximum number of perfectly distinguishable codewords are $2^5 = 32$. This proposal is proved by using a computational procedure which yields the following results.

```
=====
Total Encoded States = 64

All Encoded states are NOT mutually orthogonal.

=====
Non-Orthogonal Pairs
=====

III <-> IZZ
Overlap = (0.9999999999999996+0j)

IIX <-> IZY
Overlap = 0.9999999999999996j

IIY <-> IZX
Overlap = -0.9999999999999996j

IIZ <-> IZI
Overlap = (0.9999999999999996+0j)

IXI <-> IYZ
Overlap = 0.9999999999999996j

IXX <-> IYY
Overlap = (-0.9999999999999996+0j)

IXY <-> IYX
Overlap = (0.9999999999999996+0j)
...
ZXZ <-> ZYI
Overlap = 0.9999999999999996j

Total Non-Orthogonal Pairs = 32
```

```
=====
Orthogonal Pairs
=====
```

```
III <-> IIX
III <-> IIY
...
ZZX <-> ZZZ
ZZY <-> ZZZ
Total Orthogonal Pairs = 1984
=====
```

So, the coding results shows that amongst ${}^{64}C_2 = 2016$ possible pairs of encoded states, it was found that 1984 pairs satisfy $\langle \psi_i | \psi_j \rangle = 0$. Only 32 pairs remain non-orthogonal.

However, pairwise orthogonality alone does not guarantee that all states can simultaneously belong to the same decoding basis.

According to Graph Interpretation, the maximum-clique search on the orthogonality graph produced a clique of size 32 i.e. the largest complete subgraph (maximum clique), in which every pair of vertices is connected by an edge is 32. Hence only 32 encoded states can be perfectly distinguished using a single measurement.

This can be well visualized using the orthogonality graph encoded 64 cluster states

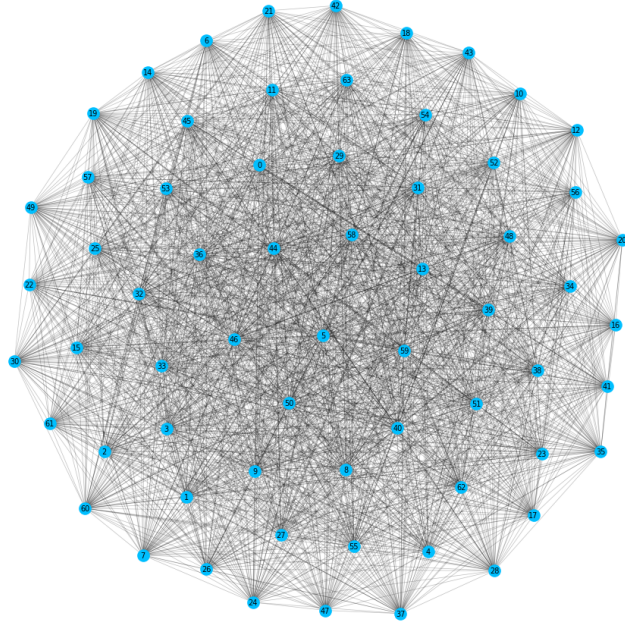


Figure 3.9: Hairball layout of 64 encoded cluster states

Alice performs those encoding operation by using pauli operators as

$$|\psi_2\rangle = (U_1 \otimes I \otimes I \otimes U_2 \otimes I \otimes U_3) |CS_6\rangle$$

where, $U_1, U_2 \in \{I, X, Y, Z\}$; $U_3 \in \{I, X\}$

The additional Z type operations on 3rd qubit of Alice does not eventually create new orthogonal states. It only changes phases within the existing subspace. That is why those operation are excluded from U_3 .

Those 32 encoding operations are shown in table below.

Table 3.4: Summary of encoding operations performed on 6 qubit cluster state by Alice

S.No.	Encoding Unitary Operator	Encoded state ($ \psi_2\rangle$)	Classical encoded message
1.	$U_0 = I \otimes I \otimes I \otimes I \otimes I \otimes I$	$\frac{1}{2}(000000\rangle + 000111\rangle + 111000\rangle + 111111\rangle)$	000000
2.	$U_1 = I \otimes I \otimes I \otimes I \otimes I \otimes X$	$\frac{1}{2}(000001\rangle + 000110\rangle + 111001\rangle + 111110\rangle)$	000001
3.	$U_2 = I \otimes I \otimes I \otimes I \otimes I \otimes Y$	$\frac{1}{2}(- 000001\rangle + 000110\rangle - 111001\rangle + 111110\rangle)$	000101
4.	$U_3 = I \otimes I \otimes I \otimes I \otimes I \otimes Z$	$\frac{1}{2}(000000\rangle - 000111\rangle + 111000\rangle - 111111\rangle)$	000100
5.	$U_4 = X \otimes I \otimes I \otimes I \otimes I \otimes I$	$\frac{1}{2}(100000\rangle + 100111\rangle + 011000\rangle + 011111\rangle)$	011000
6.	$U_5 = X \otimes I \otimes I \otimes I \otimes I \otimes X$	$\frac{1}{2}(100001\rangle + 100110\rangle + 011001\rangle + 011110\rangle)$	011001
7.	$U_6 = X \otimes I \otimes I \otimes I \otimes I \otimes Y$	$\frac{1}{2}(- 100001\rangle + 100110\rangle - 011001\rangle + 011110\rangle)$	011101
8.	$U_7 = X \otimes I \otimes I \otimes I \otimes I \otimes Z$	$\frac{1}{2}(100000\rangle - 100111\rangle + 011000\rangle - 011111\rangle)$	011101
9.	$U_8 = Y \otimes I \otimes I \otimes I \otimes I \otimes I$	$\frac{1}{2}(- 100000\rangle - 100111\rangle + 011000\rangle + 011111\rangle)$	011101
10.	$U_9 = Y \otimes I \otimes I \otimes I \otimes I \otimes X$	$\frac{1}{2}(- 100001\rangle - 100110\rangle + 011001\rangle + 011110\rangle)$	111001
11.	$U_{10} = Y \otimes I \otimes I \otimes I \otimes I \otimes Y$	$\frac{1}{2}(100001\rangle - 100110\rangle - 011001\rangle + 011110\rangle)$	111101
12.	$U_{11} = Y \otimes I \otimes I \otimes I \otimes I \otimes Z$	$\frac{1}{2}(- 100000\rangle + 100111\rangle + 011000\rangle - 011111\rangle)$	111100
13.	$U_{12} = Z \otimes I \otimes I \otimes I \otimes I \otimes I$	$\frac{1}{2}(000000\rangle + 000111\rangle - 111000\rangle - 111111\rangle)$	100000
14.	$U_{13} = Z \otimes I \otimes I \otimes I \otimes I \otimes X$	$\frac{1}{2}(000001\rangle + 000110\rangle - 111001\rangle - 111110\rangle)$	100001
15.	$U_{14} = Z \otimes I \otimes I \otimes I \otimes I \otimes Y$	$\frac{1}{2}(- 000001\rangle + 000110\rangle + 111001\rangle - 111110\rangle)$	100101
16.	$U_{15} = Z \otimes I \otimes I \otimes I \otimes I \otimes Z$	$\frac{1}{2}(000000\rangle - 000111\rangle - 111000\rangle + 111111\rangle)$	100100
17.	$U_{16} = I \otimes I \otimes I \otimes X \otimes I \otimes I$	$\frac{1}{2}(000100\rangle + 000011\rangle + 111100\rangle + 111011\rangle)$	000011
18.	$U_{17} = I \otimes I \otimes I \otimes X \otimes I \otimes X$	$\frac{1}{2}(000101\rangle + 000010\rangle + 111101\rangle + 111010\rangle)$	000010
19.	$U_{18} = I \otimes I \otimes I \otimes X \otimes I \otimes Y$	$\frac{1}{2}(- 000101\rangle + 000010\rangle - 111101\rangle + 111010\rangle)$	000110
20.	$U_{19} = I \otimes I \otimes I \otimes X \otimes I \otimes Z$	$\frac{1}{2}(000100\rangle - 000011\rangle + 111100\rangle - 111011\rangle)$	000111
21.	$U_{20} = X \otimes I \otimes I \otimes X \otimes I \otimes I$	$\frac{1}{2}(100100\rangle + 100011\rangle + 011100\rangle + 011011\rangle)$	011011
22.	$U_{21} = X \otimes I \otimes I \otimes X \otimes I \otimes X$	$\frac{1}{2}(100101\rangle + 100010\rangle + 011101\rangle + 011010\rangle)$	011010
23.	$U_{22} = X \otimes I \otimes I \otimes X \otimes I \otimes Y$	$\frac{1}{2}(- 100101\rangle + 100010\rangle - 011101\rangle + 011010\rangle)$	011110
24.	$U_{23} = X \otimes I \otimes I \otimes X \otimes I \otimes Z$	$\frac{1}{2}(100100\rangle - 100011\rangle + 011100\rangle - 011011\rangle)$	011111
25.	$U_{24} = Y \otimes I \otimes I \otimes X \otimes I \otimes I$	$\frac{1}{2}(- 100100\rangle - 100011\rangle + 011100\rangle + 011011\rangle)$	111011
26.	$U_{25} = Y \otimes I \otimes I \otimes X \otimes I \otimes X$	$\frac{1}{2}(- 100101\rangle - 100010\rangle + 011101\rangle + 011010\rangle)$	111010
27.	$U_{26} = Y \otimes I \otimes I \otimes X \otimes I \otimes Y$	$\frac{1}{2}(100101\rangle - 100010\rangle - 011101\rangle + 011010\rangle)$	111110
28.	$U_{27} = Y \otimes I \otimes I \otimes X \otimes I \otimes Z$	$\frac{1}{2}(- 100100\rangle + 100011\rangle + 011100\rangle - 011011\rangle)$	111111
29.	$U_{28} = Z \otimes I \otimes I \otimes X \otimes I \otimes I$	$\frac{1}{2}(000100\rangle + 000011\rangle - 111100\rangle - 111011\rangle)$	100011
30.	$U_{29} = Z \otimes I \otimes I \otimes X \otimes I \otimes X$	$\frac{1}{2}(000101\rangle + 000010\rangle - 111101\rangle - 111010\rangle)$	100010
31.	$U_{30} = Z \otimes I \otimes I \otimes X \otimes I \otimes Y$	$\frac{1}{2}(- 000101\rangle + 000010\rangle + 111101\rangle - 111010\rangle)$	100110
32.	$U_{31} = Z \otimes I \otimes I \otimes X \otimes I \otimes Z$	$\frac{1}{2}(000100\rangle - 000011\rangle - 111100\rangle + 111011\rangle)$	100111

All these 32 states are mutually orthogonal. After this encoding procedure the qubits which were in possession of Alice were sent to Bob via quantum channel.

The orthogonality of all these states can be visualized using the gram matrix heatmap representation

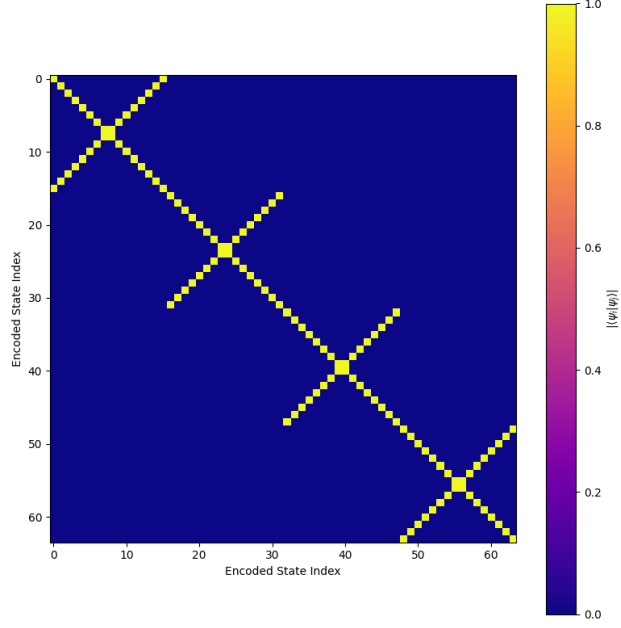


Figure 3.10: Gram matrix representation of all 64 possible encoded states

3.5.4 Decoding operations of Bob

Bob, now spontaneously applies the inverse of 6 qubit cluster state preparation operation to decode the message. The final state will be

$$|\psi_3\rangle = (H_1 \otimes I \otimes I \otimes H_2 \otimes I \otimes I) CNOT_{1,2} CNOT_{1,3} CNOT_{4,5} CNOT_{4,6} |\psi_2\rangle \quad (3.13)$$

3.5.5 Quantum Circuit

The quantum circuit consisting of the 6 qubit Cluster state preparation, encoding operations, decoding operations and measurement is shown below

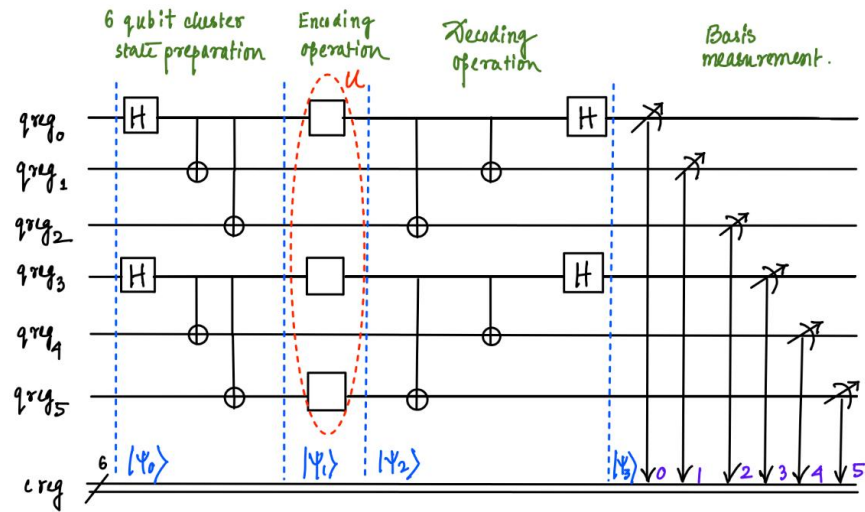
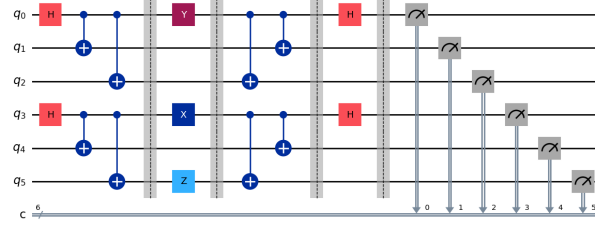


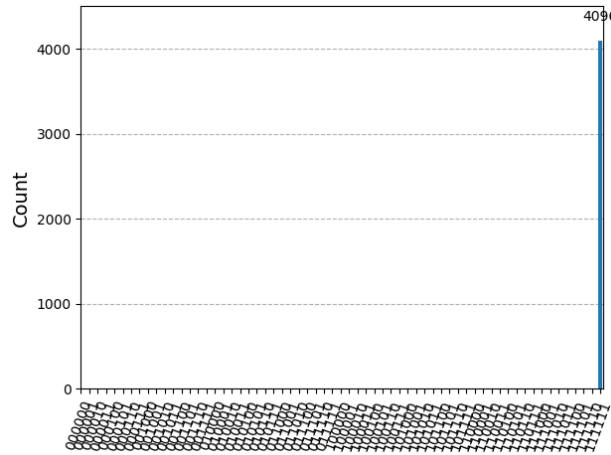
Figure 3.11: Circuit for 6 qubit Cluster state QDC

3.5.6 Implementation using Qiskit

The drawn circuit is implemented using QASM Simulator of IBM's Qiskit software for all possibilities and it is seen that for all the 32 cases the sent message can be decoded with maximal success probability. Here the corresponding circuit diagram and output result for the case no. 28 of the said scheme is shown.



(a) Circuit for 6 qubit cluster state QDC



(b) Output state after basis measurement

Figure 3.12: Implementation of 6 qubit cluster state QDC Protocol when the message 111111 is sent

3.6 Analysis of different performance metrics independent of noise

In this section, the values of different types of performance metrics, as discussed in Sec(2.4) are to be checked and reviewed. Already the performance metrics- dense coding capacity (DCC), maximum schmidt rank (Ref. Sec3) are discussed in the previous section.

This section deals with the performance metrics Holevo bound (χ) (2.4.3), channel efficiency (ζ)(2.4.5), robustness of protocol ($\bar{F}$)(2.4.6), quantum circuit depth(2.4.7), resource cost (RC)(2.4.9) for different noiseless QDC protocols.

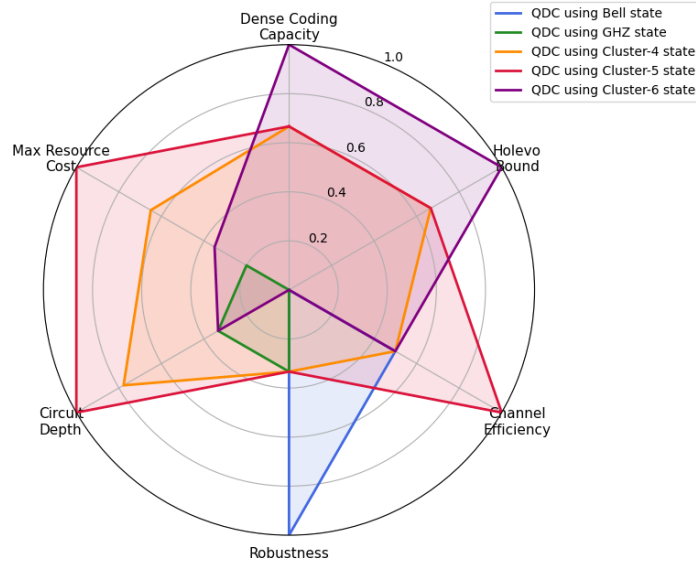
The following table summarizes the metric values obtained for different metrics for different dense coding protocols under noiseless environment.

These performance metrics can also be visualized using the performance spider chart and performance bar chart.

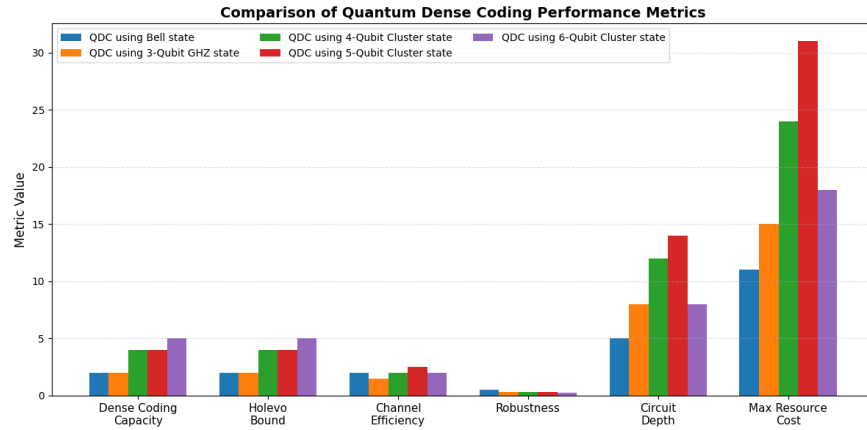
Table 3.5: Comparison of different performance metrics for different quantum dense coding protocols under noiseless environment.

Employed Protocol	Holevo bound	Channel Efficiency	Robustness	Circuit Depth	Resource Cost
Bell State	2	2.00	0.50	5	11
3-Qubit GHZ	2	1.50	0.33	8	15
4-Qubit Cluster	4	2.00	0.33	12	24
5-Qubit Cluster	4	2.50	0.33	14	31
6-Qubit Cluster	5	2.00	0.25	8	18

Comparative Performance Analysis of Quantum Dense Coding Protocols



(a) Spider chart



(b) Bar chart

Figure 3.13: Comparison of performance metrics of different entangled channel QDC protocols

Chapter 4

Quantum dense coding protocols in noisy environment

In the previous chapter, quantum dense coding protocols employing Bell, GHZ, and multiqubit cluster-state quantum channels were investigated under ideal noiseless conditions, where perfect state preparation, transmission, and decoding were assumed. However, practical quantum communication systems inevitably suffer from environmental interactions and hardware imperfections that introduce errors during the transmission of quantum information. Consequently, the performance of a dense coding protocol deteriorates as the transmitted qubits undergo decoherence and various quantum noise processes.

This chapter investigates the robustness of the proposed quantum dense coding protocols under realistic noisy environments by incorporating three fundamental Pauli error models, namely the bit-flip, phase-flip, and bit-phase-flip channels. For each noise model, the Bell-state, three-qubit GHZ-state, and four-, five-, and six-qubit cluster-state dense coding protocols are implemented using the Qiskit framework. Their performances are then evaluated and compared in terms of output state fidelity, dense coding capacity, channel efficiency, quantum bit error rate, and quantum resource cost. The results obtained in this chapter provide a quantitative understanding of the influence of quantum noise on different entangled quantum channels and help identify the most robust channel for quantum dense coding applications.

4.1 QDC over bit flip channel

The bit-flip channel represents one of the simplest and most widely studied quantum noise models. It is the quantum analogue of the classical binary symmetric channel, where the logical states $|0\rangle$ and $|1\rangle$ are interchanged with probability η_B . Since the transmitted qubits are susceptible to such errors during communication, analysing the effect of bit-flip noise provides valuable insight into the robustness of quantum dense coding protocols. In this section, the bit-flip channel is incorporated into each dense coding protocol developed in the previous chapter, and their performances are systematically compared under identical noise conditions.

4.1.1 Implementation of QDC Protocol using Bell state

The quantum dense coding protocol using Bell-state discussed in Chapter 3 (Sec. 3.1) is investigated here in the presence of bit-flip noise. After Alice performs the encoding operation, the transmitted qubit is assumed to pass through a bit-flip quantum channel before reaching Bob. The resulting noisy quantum state is then decoded using the decoding operation circuit. By varying the bit-flip probability from $\eta_B = 0$ to $\eta_B = 1$, the output state fidelity is computed to evaluate the degradation of the communication protocol caused by quantum noise. The protocol under bit flip noise is discussed below.

The density matrix of the original state before encoding

$$\rho = |\psi_1\rangle \langle \psi_1| = |\phi^+\rangle \langle \phi^+|$$

Suppose, the encoding operator is $(X \otimes I)$ and the corresponding classical message sent is 01 (Ref. Sec.3.1.3) then

$$\begin{aligned}\rho_{enc}^{01} &= (X \otimes I) |\phi^+\rangle \langle \phi^+| (X \otimes I)^\dagger \\ \rho_{enc}^{01} &= |\psi^+\rangle \langle \psi^+|\end{aligned}$$

The qubit is now send to Bob via transmission channels where applied noise effects can affect the qubit state. Therefore, the density matrix becomes

$$\rho_{noise_B}^{01} = (1 - \eta_B) |\psi^+\rangle \langle \psi^+| + \eta_B |\phi^+\rangle \langle \phi^+|$$

As both the qubits are now in possession of Bob and he performs decoding operation on them so that the density matrix reduces to

$$\rho_{out_B}^{01} = (1 - \eta_B) |01\rangle \langle 01| + \eta_B |00\rangle \langle 00|$$

and,

$$\rho_{out_B}^{01} = \begin{bmatrix} \eta_B & 0 & 0 & 0 \\ 0 & (1-\eta_B) & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore, the fidelity to decode the correct state $|01\rangle$ will be

$$\begin{aligned}F_B^m &= \langle \psi_{ideal} | \rho_{out}^m | \psi_{ideal} \rangle = \langle 01 | \rho_{out_B}^{01} | 01 \rangle \\ F_B^{01} &= 1 - \eta_B\end{aligned}$$

This proposal is implemented using QASM simulator and for $\eta_B = 0.1, 0.5$ and 0.9 the output fidelity is shown below along with the implementation circuit.

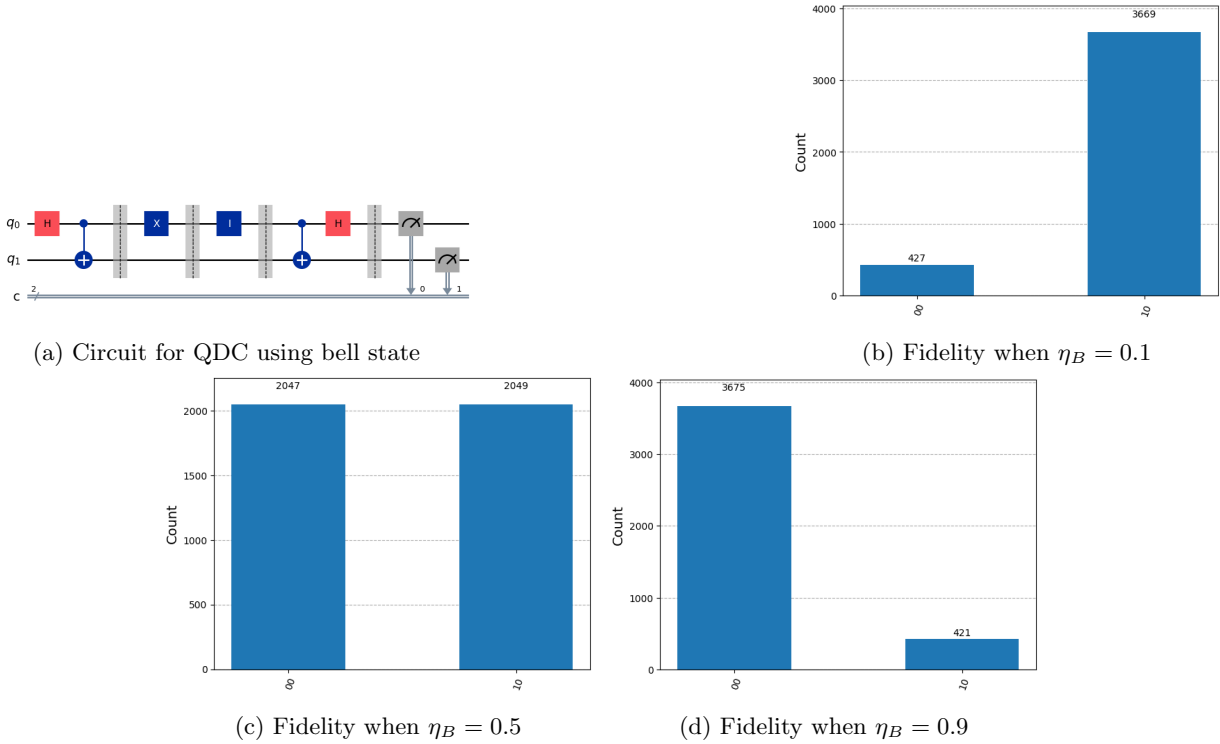


Figure 4.1: Implementation of QDC Protocol using bell state when the message 01 is sent through bit flip channel

4.1.2 Implementation of QDC Protocol using 3 qubit GHZ state

The GHZ-state dense coding protocol using three-qubit is next analysed under the bit-flip noise model. In this protocol, the encoded qubits transmitted to Bob are subjected to independent bit-flip errors before reaching Bob, same as like the previous case. The decoded output states are compared with the corresponding ideal states to quantify the influence of noise on multipartite entanglement. The results are subsequently analysed using the performance metric, Fidelity, introduced in Chapter 2. The density matrix of the original state before encoding

$$\rho = |GHZ\rangle \langle GHZ|$$

Suppose the encoding operator is $(X \otimes I)$ and the classical encoded message is 011 (Ref. table3.1) then

$$\rho_{enc}^{011} = \frac{1}{2}(|100\rangle \langle 100| + |100\rangle \langle 011| + |011\rangle \langle 100| + |011\rangle \langle 011|)$$

The qubits are now send to Bob via transmission channels where noise effects can alter the qubit state. Therefore, applying the bit flip noise model the density matrix becomes

$$\rho_{noise_B}^{011} = (1 - \eta_B)^2 |\psi_1\rangle \langle \psi_1| + \eta_B(1 - \eta_B) |\psi_2\rangle \langle \psi_2| + \eta_B(1 - \eta_B) |\psi_3\rangle \langle \psi_3| + \eta_B^2 |\psi_4\rangle \langle \psi_4|$$

where, the states

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|100\rangle + |011\rangle)$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

$$|\psi_3\rangle = \frac{1}{\sqrt{2}}(|110\rangle + |001\rangle)$$

$$|\psi_4\rangle = \frac{1}{\sqrt{2}}(|010\rangle + |101\rangle)$$

and are transformed using 4 two-qubit Kraus operators ($K_{ij} = A_i \otimes A_j$) (Ref. 2.29,2.30)

$$K_{00} = (1 - \eta_B)^2 (I \otimes I \otimes I)$$

$$K_{01} = \eta_B(1 - \eta_B) (X \otimes I \otimes I)$$

$$K_{10} = \eta_B(1 - \eta_B) (I \otimes X \otimes I)$$

$$K_{11} = \eta_B^2 (X \otimes X \otimes I)$$

As all the qubits are now in possession of Bob and he performs decoding operation on them so that the density matrix reduces to

$$\rho_{out_B}^{011} = (1 - \eta_B)^2 |011\rangle \langle 011| + \eta_B(1 - \eta_B) |000\rangle \langle 000| + \eta_B(1 - \eta_B) |001\rangle \langle 001| + \eta_B^2 |010\rangle \langle 010|$$

Therefore, the fidelity to decode the correct state $|011\rangle$ will be

$$F_B^m = \langle \psi_{ideal} | \rho_{out_B}^m | \psi_{ideal} \rangle = \langle 011 | \rho_{out_B}^{011} | 011 \rangle$$

$$F_B^{01} = (1 - \eta_B)^2$$

This proposal is implemented using QASM simulator and for $\eta_B = 0.1, 0.5$ and 0.9 the output fidelity is shown below along with the implementation circuit.

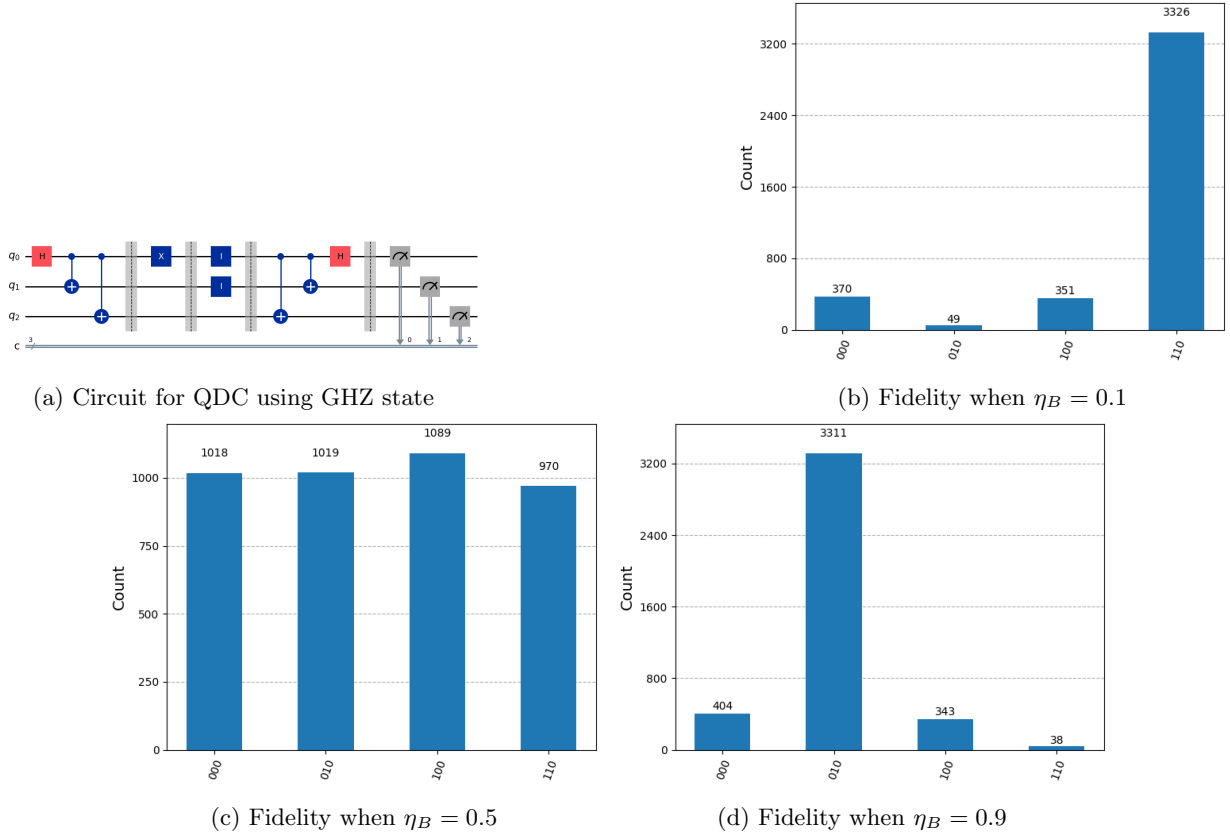


Figure 4.2: Implementation of QDC Protocol using GHZ state when the message 011 is sent through bit flip channel

Remark. The construction of the four- and five-qubit cluster-state quantum dense coding protocols follows the same methodology employed for the Bell-state and GHZ-state protocols, involving state preparation, local unitary encoding, orthogonality analysis, and decoding. Presenting the complete mathematical derivations for these intermediate cluster-state channels would largely repeat the same sequence of calculations while increasing the length of the report considerably. Therefore, only the final simulation results are presented for these protocols.

4.1.3 Implementation of QDC protocol using 4- and 5-qubit cluster state

The QDC protocol using 4-qubit cluster state under bit flip noise is employed in Qiskit where the classical message 0110 is sent (Ref. Table 3.2). The corresponding circuit and simulation results for bit flip noise $\eta_B = 0.1, 0.5$ and 0.9 are shown below.

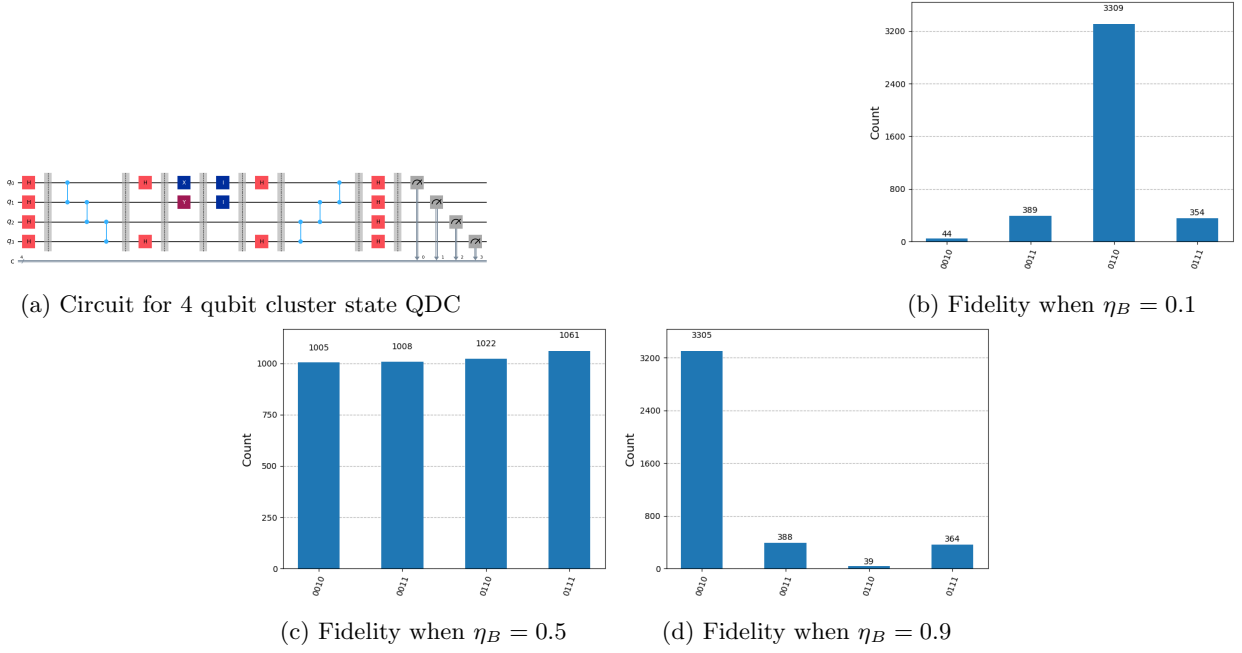


Figure 4.3: Implementation of QDC Protocol using 4 qubit cluster state when the message 0110 is sent through bit flip channel

Similarly, the QDC protocol using 5-qubit cluster state under bit flip noise is also employed in Qiskit where the classical message 01100 (Ref. Table 3.3) is sent. The corresponding circuit and simulation results for bit flip noise $\eta_B = 0.1, 0.5$ and 0.9 are shown below.

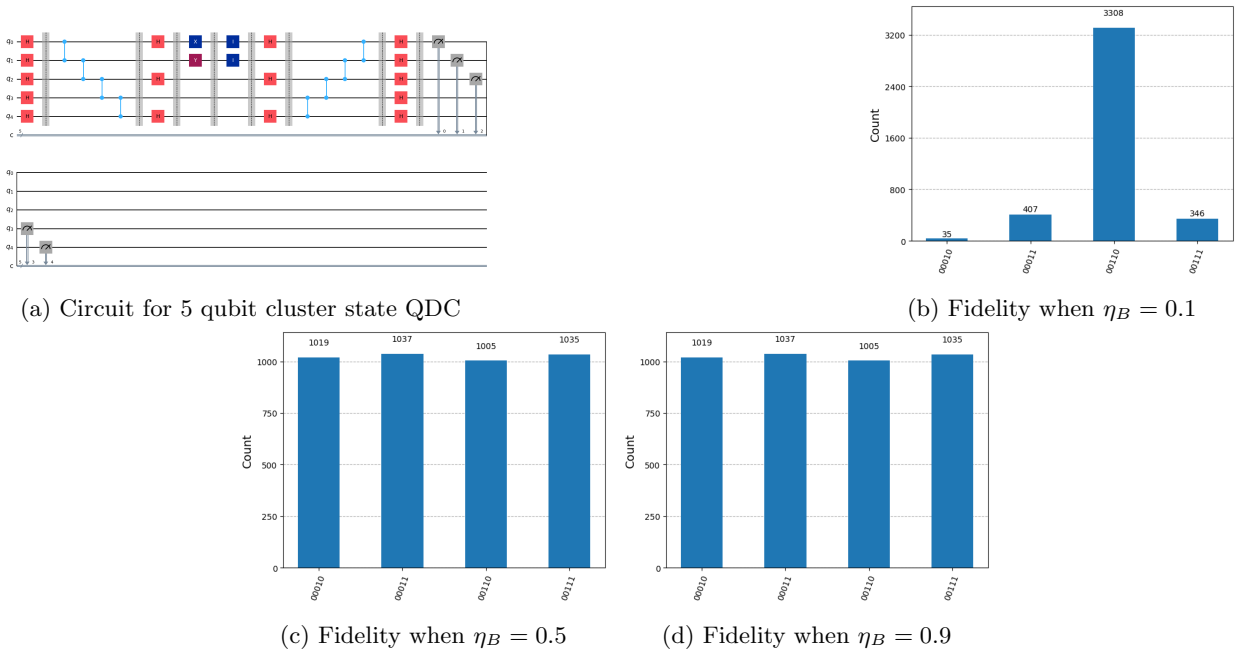


Figure 4.4: Implementation of QDC Protocol using 5 qubit cluster state when the message 01100 is sent through bit flip channel

Remark. The six-qubit cluster-state quantum channel is discussed in greater detail as it represents the most general dense coding protocol, where three qubits carries the encoded message. Unlike the four- and five-qubit cases, the six-qubit CS protocol exhibits a substantially larger encoding space, requiring a complex orthogonality analysis to identify the maximum set of mutually orthogonal codewords.

4.1.4 Implementation of QDC Protocol using 6 qubit cluster state

Finally, the dense coding protocol using six-qubit cluster-state is analysed under the bit-flip noise model. The transmitted qubits are subjected to bit-flip errors prior to decoding, and the protocol performance is evaluated by analysing the output fidelity as functions of the bit-flip probability. The obtained results are then compared with those of the Bell, GHZ, and lower-dimensional cluster-state protocols to assess the influence of increasing system size on noise robustness. The density matrix of the original state before encoding

$$\rho = |CS_6\rangle \langle CS_6|$$

Suppose the encoding operator is $(X \otimes I \otimes I \otimes I \otimes I \otimes X)$ and the classical encoded message is 011001 (Ref. Table3.4) then

$$\rho_{enc}^{011001} = \langle \psi_{enc} | \psi_{enc} \rangle$$

where,

$$|\psi_{enc}\rangle = \frac{1}{2}(|100001\rangle + |100110\rangle + |011001\rangle + |011110\rangle)$$

Alice now sends the qubits in her possession to Bob via transmission channels where noise models can affect the qubit state. Therefore, applying the bit flip noise model the density matrix becomes

$$\begin{aligned} \rho_{noise_B}^{011001} &= (1 - \eta_B)^3 |\psi_1\rangle \langle \psi_1| + \eta_B(1 - \eta_B)^2 \left(|\psi_2\rangle \langle \psi_2| + |\psi_3\rangle \langle \psi_3| + |\psi_5\rangle \langle \psi_5| \right) \\ &+ \eta_B^2(1 - \eta_B) \left(|\psi_4\rangle \langle \psi_4| + |\psi_6\rangle \langle \psi_6| + |\psi_7\rangle \langle \psi_7| \right) + \eta_B^3 |\psi_8\rangle \langle \psi_8|. \end{aligned}$$

where, the states

$$\begin{aligned} |\psi_1\rangle &= \frac{1}{2}(|100001\rangle + |100110\rangle + |011001\rangle + |011110\rangle) \\ |\psi_2\rangle &= \frac{1}{2}(|100000\rangle + |100111\rangle + |011000\rangle + |011111\rangle) \\ |\psi_3\rangle &= \frac{1}{2}(|100101\rangle + |100010\rangle + |011101\rangle + |011010\rangle) \\ |\psi_4\rangle &= \frac{1}{2}(|100100\rangle + |100011\rangle + |011100\rangle + |011011\rangle) \\ |\psi_5\rangle &= \frac{1}{2}(|000001\rangle + |000110\rangle + |111001\rangle + |111110\rangle) \\ |\psi_6\rangle &= \frac{1}{2}(|000000\rangle + |000111\rangle + |111000\rangle + |111111\rangle) \\ |\psi_7\rangle &= \frac{1}{2}(|000101\rangle + |000010\rangle + |111101\rangle + |111010\rangle) \\ |\psi_8\rangle &= \frac{1}{2}(|000100\rangle + |000011\rangle + |111100\rangle + |111011\rangle) \end{aligned}$$

and are transformed using eight, three-qubit Kraus operators ($K_{ijl} = A_i \otimes A_j \otimes A_l$) (Ref. 2.29,2.30)

$$\begin{aligned} K_{000} &= (1 - \eta_B)^3 (I_1 \otimes I_4 \otimes I_6) \\ K_{001} &= \eta_B(1 - \eta_B)^2 (I_1 \otimes I_4 \otimes X_6) \\ K_{010} &= \eta_B(1 - \eta_B)^2 (I_1 \otimes X_4 \otimes I_6) \end{aligned}$$

$$K_{011} = \eta_B^2(1 - \eta_B)(I_1 \otimes X_4 \otimes X_6)$$

$$K_{100} = \eta_B(1 - \eta_B)^2(X_1 \otimes I_4 \otimes I_6)$$

$$K_{101} = \eta_B^2(1 - \eta_B)(X_1 \otimes I_4 \otimes X_6)$$

$$K_{110} = \eta_B^2(1 - \eta_B)(X_1 \otimes X_4 \otimes I_6)$$

$$K_{111} = \eta_B^3(X_1 \otimes X_4 \otimes X_6)$$

As all the qubits are now in possession of Bob and he performs decoding operation on them so that the density matrix reduces to

$$\rho_{out_B}^{011001} = U_{dec} \rho_{noise_B} U_{dec}^\dagger$$

Therefore, the fidelity to decode the correct state $|011001\rangle$ will be

$$F_B^m = \langle \psi_{ideal} | \rho_{out_B}^m | \psi_{ideal} \rangle = \langle 011001 | \rho_{out_B}^{011001} | 011001 \rangle$$

$$F^{01} = (1 - \eta_B)^3$$

This proposal is implemented using QASM simulator and for $\eta_B = 0.1$ and 0.5 the output fidelity is shown below along with the implementation circuit.

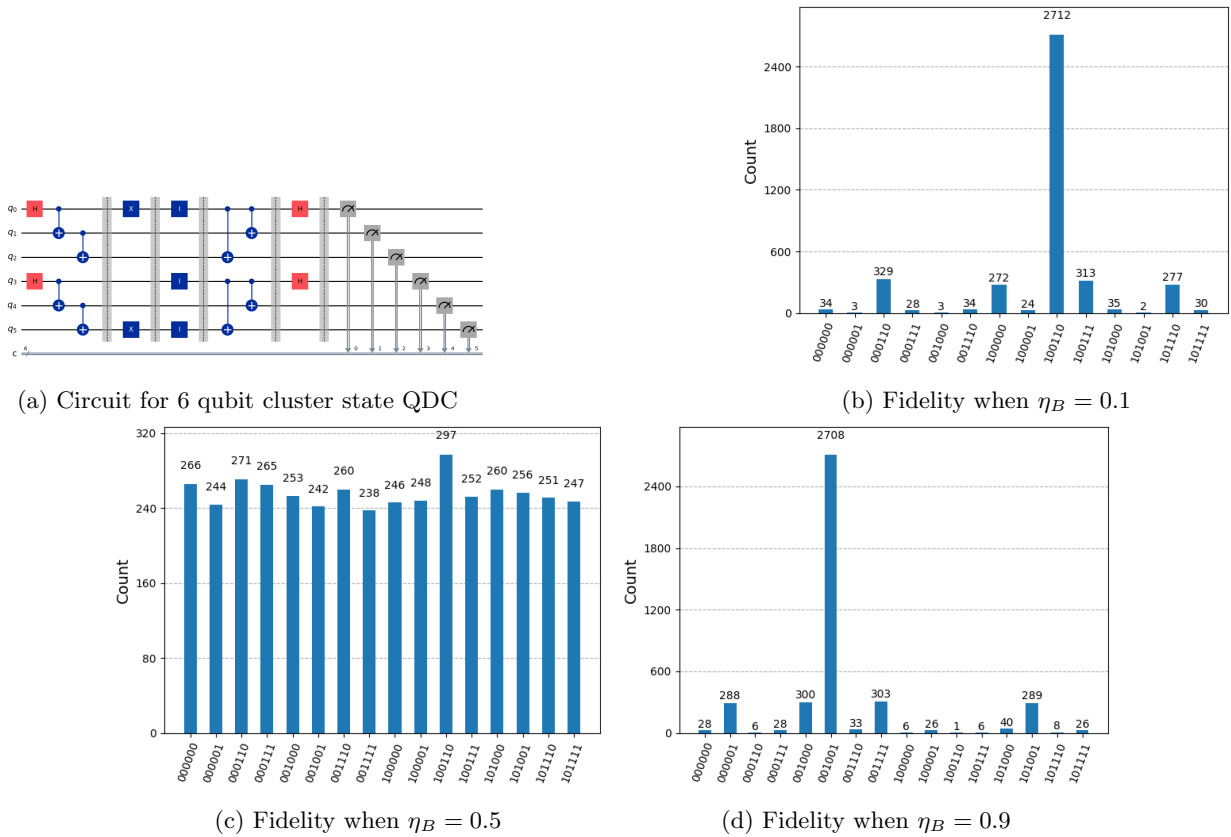
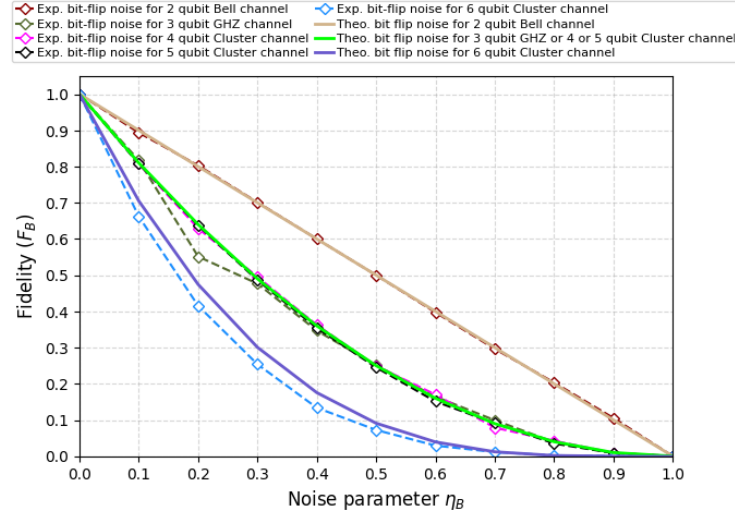


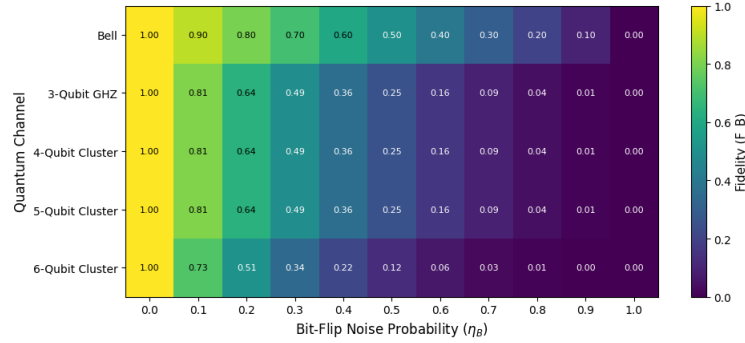
Figure 4.5: Implementation of QDC Protocol using 6 qubit cluster state when the message 011001 is sent through bit flip channel

To compare the robustness of the various entangled quantum channels against bit-flip noise, the output state fidelity is evaluated as a function of the bit-flip probability η_B . The fidelity quantifies the exactness between the ideal decoded state and the state obtained after transmission through the noisy quantum

channel. The following plot illustrates the variation of the output state fidelity for the Bell-state, three-qubit GHZ-state, and four-, five-, six-qubit cluster-state dense coding protocols with respect to the noise parameter.



(a) Variation of Fidelity(F) against noise parameter(η_B)



(b) Heatmap representation of theoretical fidelity variation against noise parameter

Figure 4.6: Comparison of robustness of different entangled channel QDC protocols under bit flip noise

4.2 QDC over phase flip channel

The phase-flip channel is another fundamental Pauli noise model that describes errors affecting the relative phase of a quantum state without altering its computational basis states. Unlike the bit-flip channel, phase-flip noise preserves the logical values of the qubits while introducing phase reversals that degrade quantum coherence and entanglement. Since quantum dense coding relies heavily on the preservation of entanglement, phase-flip errors can significantly influence the decoding accuracy and communication performance. In this section the quantum dense coding protocols are investigated under the phase-flip noise model.

4.2.1 Implementation of QDC Protocol using Bell state

Here, the quantum dense coding protocol using Bell state is analysed under the phase-flip noise model. Following the encoding process, the transmitted qubit is assumed to undergo phase-flip errors before reaching Bob. The decoding operation is then performed using the inverse Bell-state preparation circuit, and the recovered quantum state is compared with the ideal state. The protocol is simulated for different phase-flip

probabilities to evaluate the degradation in communication performance caused by phase errors. The protocol simply follows the process as discussed in Sec (4.1.1) with a change in the implemented noise model. The Kraus operators used in this case corresponding to the phase flip noise are mentioned previously (Ref. eq2.32,2.33). Therefore the output density matrix for the encoded message 01 (Ref. Sec. 3.1.3) will be

$$\rho_{out_W}^{01} = (1 - \eta_W) |01\rangle \langle 01| + \eta_W |11\rangle \langle 11|$$

and state fidelity

$$F_W^{01} = 1 - \eta_W$$

The corresponding simulation results are provided below.

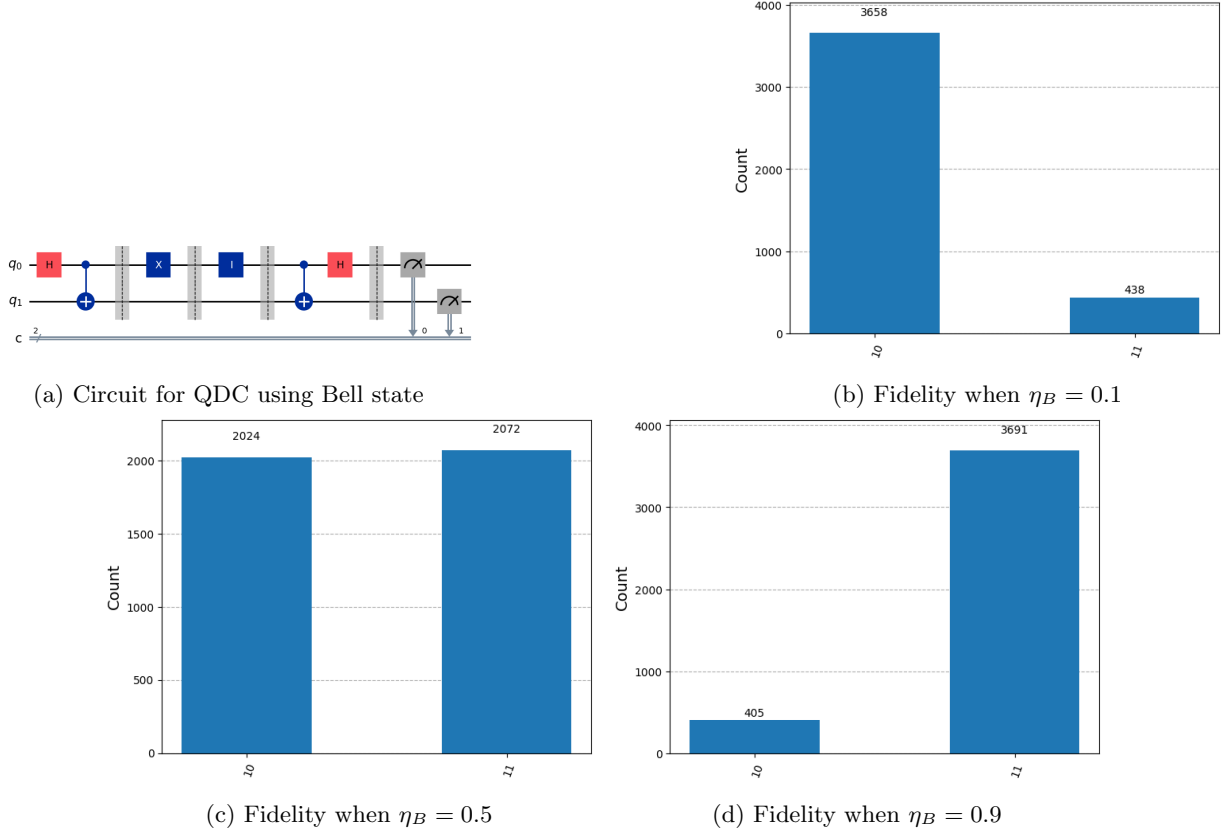


Figure 4.7: Implementation of QDC Protocol using Bell state when the message 01 is sent through phase flip channel

4.2.2 Implementation of QDC Protocol using 3 qubit GHZ state

The quantum dense coding protocol using GHZ-state is analysed next, under the phase-flip noise model. Following the encoding process, the transmitted qubits are assumed to undergo phase-flip errors before reaching Bob, similar to that in case of Bell state. The decoding operation is then performed and the recovered quantum state is compared with the ideal state. The influence of phase decoherence on the multipartite entanglement is analysed by comparing the decoded output state with the corresponding ideal state over a range of phase-flip probabilities.

The protocol simply follows the process as discussed in Sec (4.1.2) with a change in the implemented noise model. The two qubit Kraus operators, constituting independent single qubit counterparts (Ref. 2.32, 2.33), corresponding to the phase flip noise are

$$K_{00} = (1 - \eta_W)^2 (I \otimes I \otimes I)$$

$$K_{01} = \eta_W(1 - \eta_W)(Z \otimes I \otimes I)$$

$$K_{10} = \eta_W(1 - \eta_W)(I \otimes Z \otimes I)$$

$$K_{11} = \eta_W^2(Z \otimes Z \otimes I)$$

and the final density matrix for the encoded message 011 (Ref. Table. 3.1) is

$$\rho_W^{011} = \frac{1 + (1 - 2\eta_W)^2}{2} |011\rangle \langle 011| + \frac{1 - (1 - 2\eta_W)^2}{2} |111\rangle \langle 111|$$

Therefore the state fidelity is

$$F_{out_W}^{011} = \frac{1 + (1 - 2\eta_W)^2}{2} = 1 - 2\eta_W + 2\eta_W^2$$

The corresponding Qiskit simulation are given below.

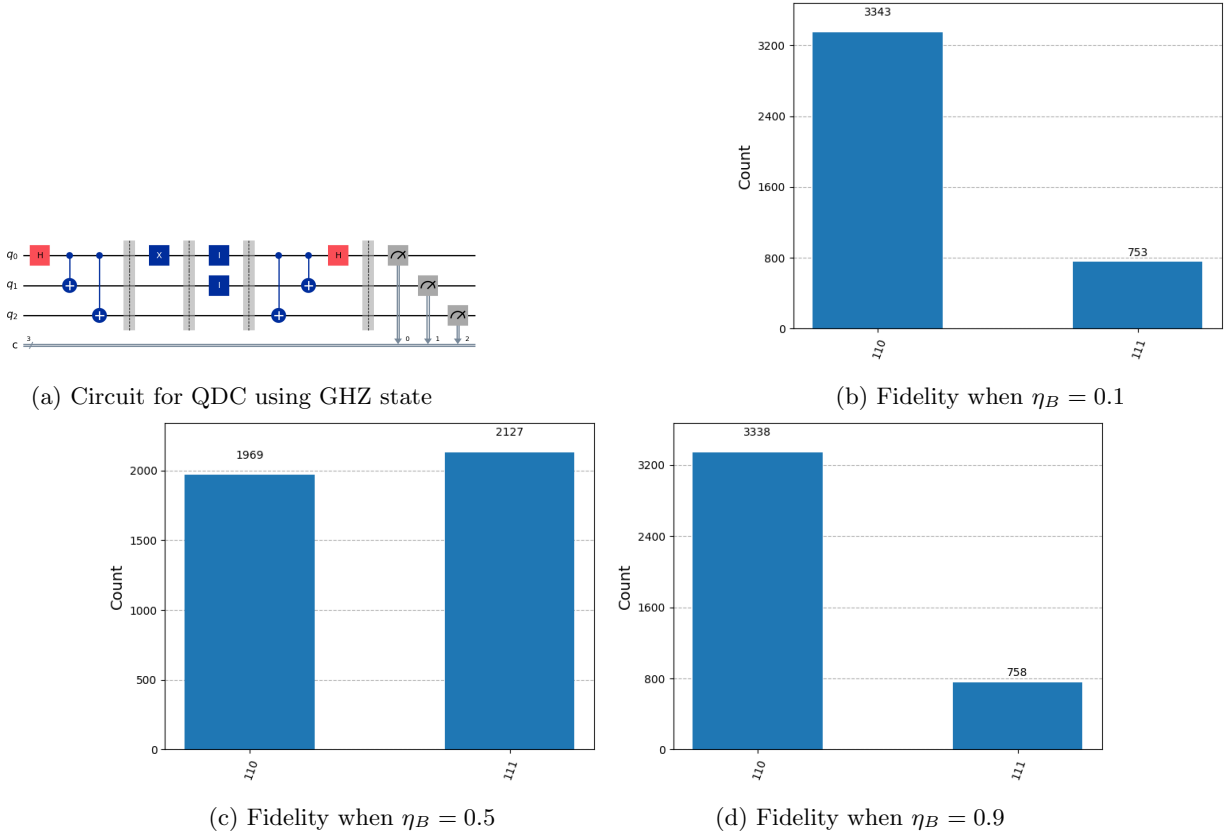


Figure 4.8: Implementation of QDC Protocol using GHZ state when the message 011 is sent through phase flip channel

Remark. Again, the construction of the four- and five-qubit cluster-state quantum dense coding protocols follows the same methodology employed for the Bell-state and GHZ-state protocols. Therefore for the sake of brevity the complete mathematical derivations of QDC in phase flip models for these intermediate cluster-state channels are omitted out.

4.2.3 Implementation of QDC Protocol using 4- and 5- qubit cluster state

The QDC protocol using 4-qubit cluster state under phase flip noise is employed in Qiskit where the classical message 0110 is sent (Ref. Table 3.2). The corresponding circuit and simulation results for bit flip noise $\eta_B = 0.1, 0.5$ and 0.9 are shown below.

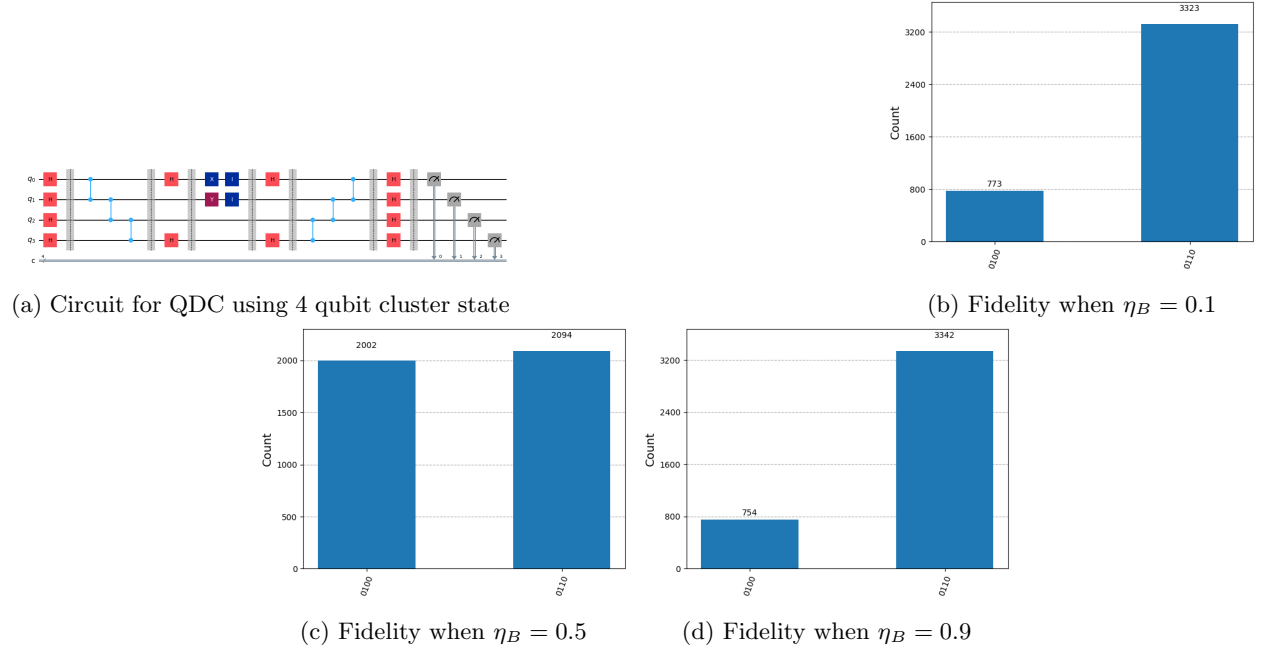


Figure 4.9: Implementation of QDC Protocol using 4 qubit cluster state when the message 0110 is sent through phase flip channel

Likewise, the QDC protocol using the 5-qubit cluster state under phase flip noise is also employed in Qiskit where the classical message 01100 is sent (Ref. Table 3.3). The corresponding circuit and simulation results for bit flip noise $\eta_B = 0.1, 0.5$ and 0.9 are shown below.

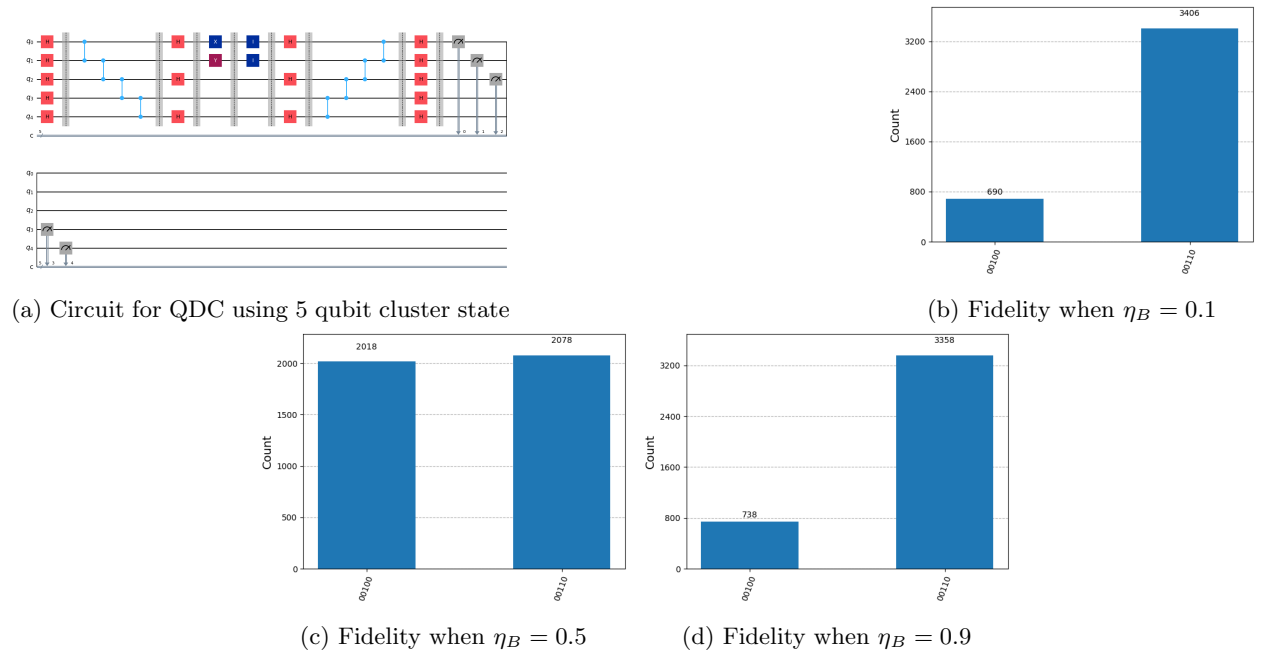


Figure 4.10: Implementation of QDC Protocol using 5 qubit cluster state when the message 01100 is sent through phase flip channel

4.2.4 Implementation of QDC Protocol using 6 qubit cluster state

Finally, the dense coding protocol using six-qubit cluster-state is implemented under the phase-flip noise model. The transmitted qubit is assumed to undergo phase-flip errors before reaching Bob. After the decoding process, the recovered quantum state is compared with the ideal state like before. The protocol is simulated for different phase-flip probabilities to evaluate the degradation in communication performance caused by phase errors.

The protocol simply follows the process as discussed in Sec (4.1.4) with a change in the implemented noise model. The Kraus operators corresponding to the phase flip noise in this case are

$$\begin{aligned} K_{000} &= (1 - \eta_W)^3 (I_1 \otimes I_4 \otimes I_6) \\ K_{001} &= \eta_W (1 - \eta_W)^2 (I_1 \otimes I_4 \otimes Z_6) \\ K_{010} &= \eta_W (1 - \eta_W)^2 (I_1 \otimes Z_4 \otimes I_6) \\ K_{011} &= \eta_W^2 (1 - \eta_W) (I_1 \otimes Z_4 \otimes Z_6) \\ K_{100} &= \eta_W (1 - \eta_W)^2 (Z_1 \otimes I_4 \otimes I_6) \\ K_{101} &= \eta_W^2 (1 - \eta_W) (Z_1 \otimes I_4 \otimes Z_6) \\ K_{110} &= \eta_W^2 (1 - \eta_W) (Z_1 \otimes Z_4 \otimes I_6) \\ K_{111} &= \eta_W^3 (Z_1 \otimes Z_4 \otimes Z_6) \end{aligned}$$

Using this operators the protocol is developed and implemented using QASM simulator, again and for $\eta_B = 0.1, 0.5$ and 0.9 the output fidelity is shown below. Here classical message 011001 is sent (Ref. Table 3.4).

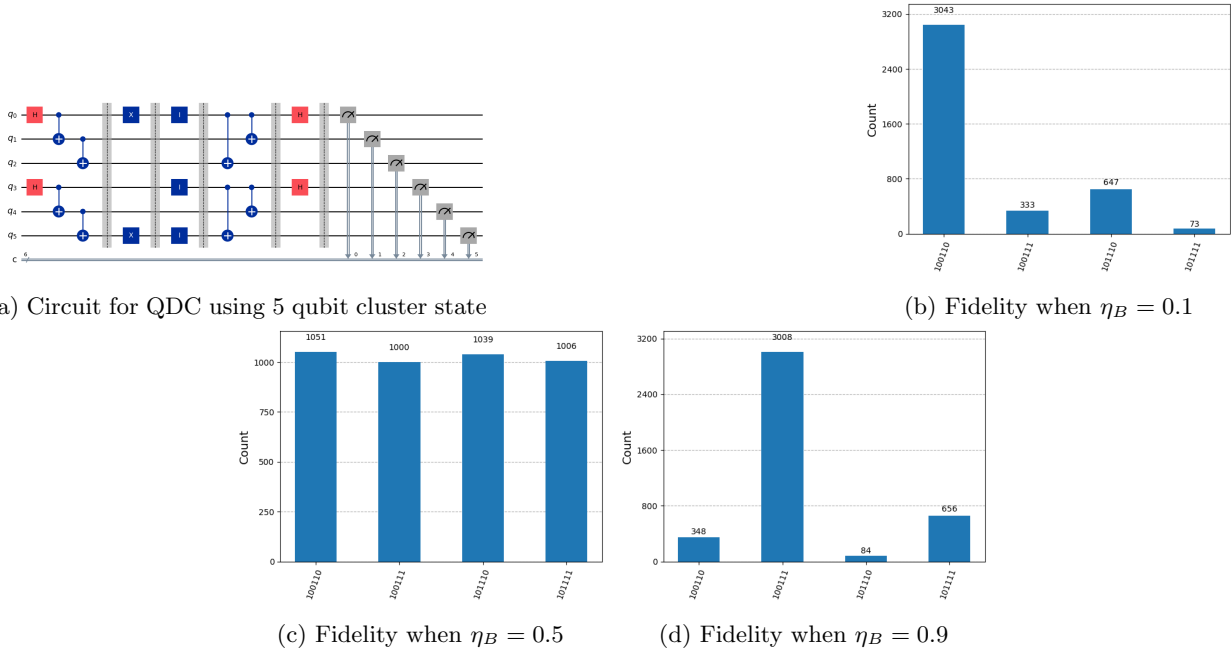
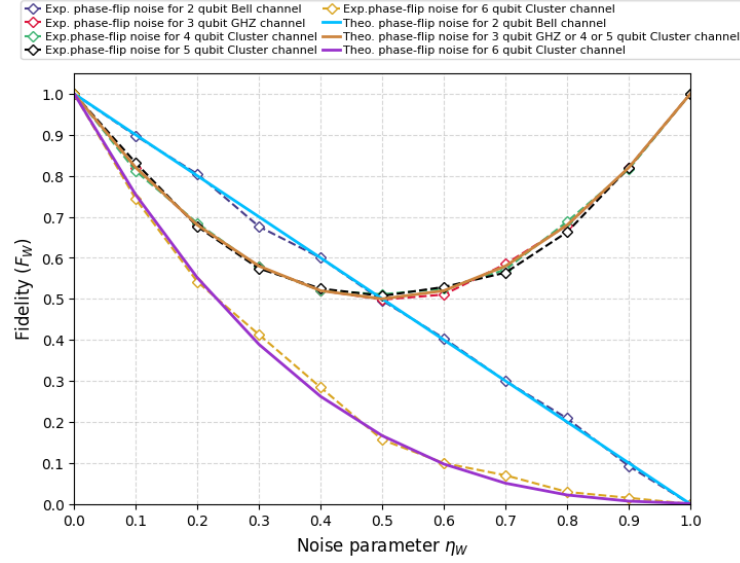
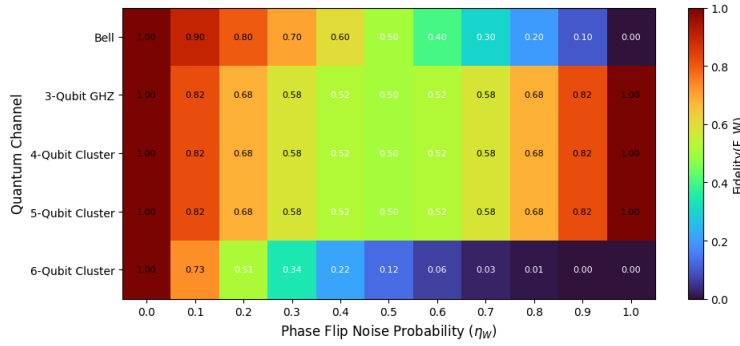


Figure 4.11: Implementation of QDC Protocol using 6 qubit cluster state when the message 011001 is sent through phase flip channel

To check the robustness of all the abovementioned QDC protocol using various entangled quantum channels against phase-flip noise, the output state fidelity is evaluated as a function of the phase-flip probability η_W . The following plot illustrates the same.

(a) Variation of Fidelity(F) against noise parameter(η_W)

(b) Heatmap representation of theoretical fidelity variation against noise parameter

Figure 4.12: Comparison of robustness of different entangled channel QDC protocols under phase flip noise

4.3 QDC over bit-phase flip channel

The bit-phase flip channel represents a Pauli noise model in which both bit-flip and phase-flip errors occur simultaneously. As a result, it generally introduces more severe degradation in quantum communication protocols than either the bit-flip or phase-flip channel individually. In this section, the entangled-state quantum dense coding protocols are implemented under the bit-phase flip noise model. Their performances are analysed and compared using the output state fidelity introduced in Chapter 2.

4.3.1 Implementation of QDC Protocol using Bell state

The quantum dense coding protocol using Bell-state is first investigated under the bit-phase flip noise model. After Alice encodes the classical message, the transmitted qubit is subjected to bit-phase flip errors before reaching Bob. The decoded output state is then compared with the corresponding ideal state to evaluate the effect of the combined Pauli errors on the communication performance.

The protocol simply follows the process as discussed in Sec (4.1.1) with a change in the implemented noise model. The single qubit Kraus operators corresponding to the bit-phase flip noise are used here (Ref. eq.2.35,

2.36). Therefore the state fidelity, when say encoded message sent is 01 (Ref. Sec3.1.3) becomes

$$F_F^{01} = 1 - \eta_F$$

The corresponding simulation results are provided below.

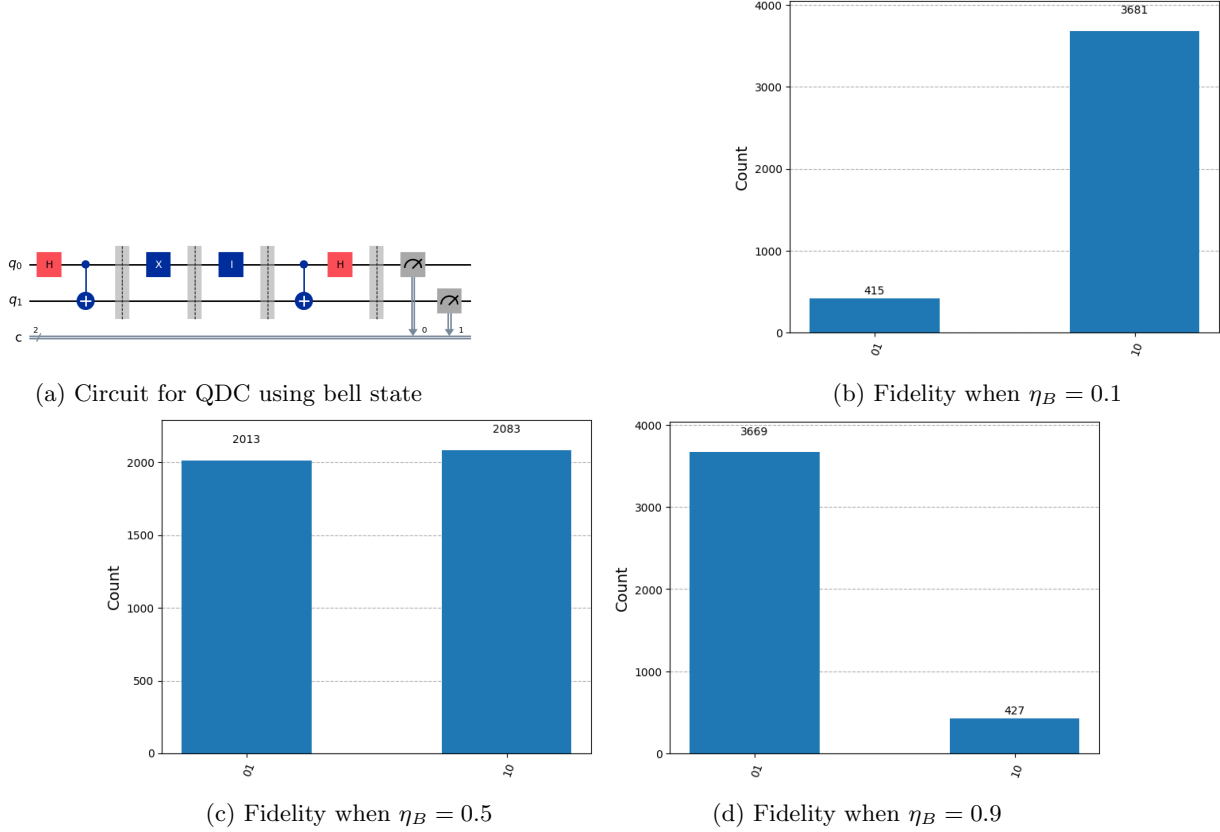


Figure 4.13: Implementation of QDC Protocol using bell state when the message 01 is sent through bit-phase flip channel

4.3.2 Implementation of QDC Protocol using 3 qubit GHZ state

The dense coding protocol using three-qubit GHZ-state is next analysed in the presence of bit-phase flip noise. The encoded qubits transmitted by Alice undergo independent bit-phase flip errors before the decoding operation is performed by Bob. The influence of the noise on the multipartite entangled channel is examined by evaluating the output state fidelity for different noise probabilities.

The protocol simply follows the process as discussed in Sec (4.1.2) obviously, with a change in the implemented noise model. The Kraus operators corresponding to the bit-phase flip noise are

$$K_{00} = (1 - \eta_F)^2 (I \otimes I \otimes I)$$

$$K_{01} = \eta_F (1 - \eta_F) (Y \otimes I \otimes I)$$

$$K_{10} = \eta_F (1 - \eta_F) (I \otimes Y \otimes I)$$

$$K_{11} = \eta_F^2 (Y \otimes Y \otimes I)$$

and the final state fidelity corresponding to say encoded message 011 (Ref. Table 3.1) is

$$F_{out_F}^{011} = (1 - \eta_F)^2$$

The corresponding Qiskit simulation are given below.

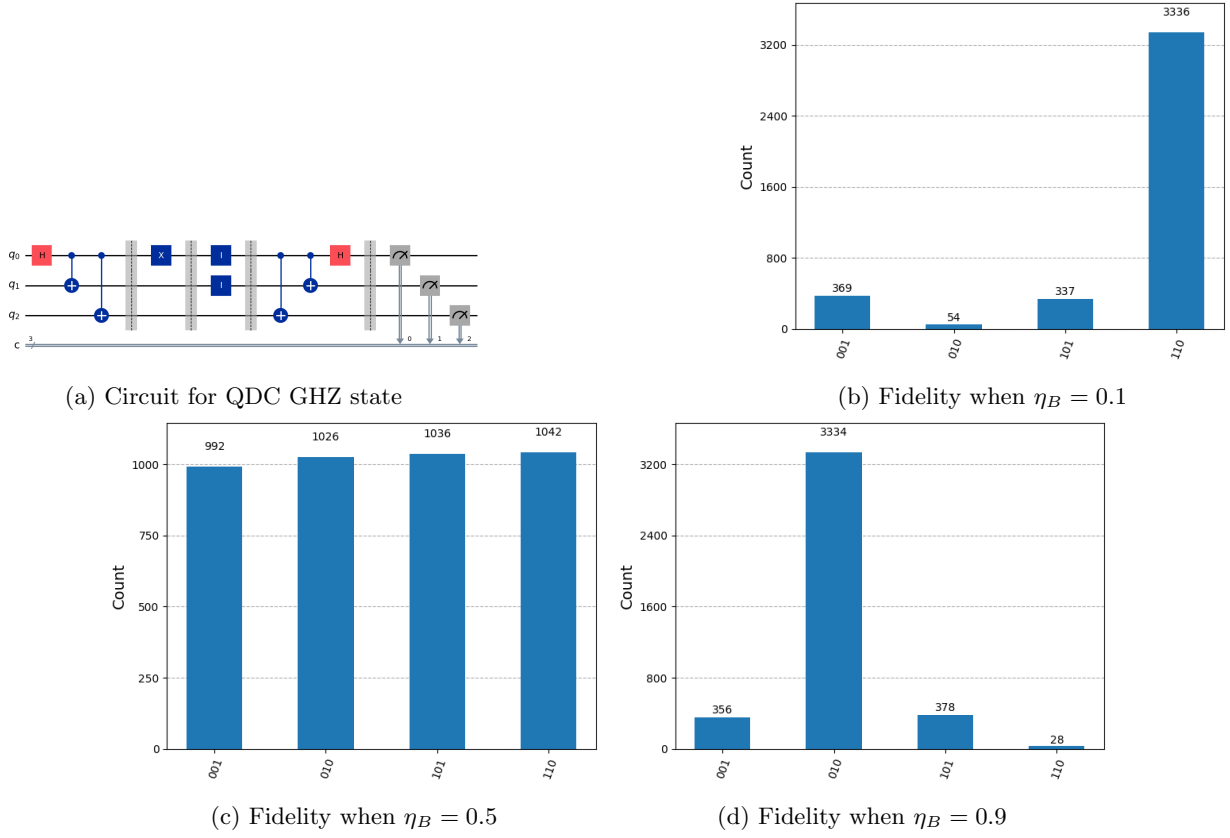


Figure 4.14: Implementation of QDC Protocol using GHZ state when the message 011 is sent through bit-phase flip channel

4.3.3 Implementation of QDC Protocol using 4- and 5- qubit cluster state

The QDC protocol using 4-qubit cluster state under bit-phase flip noise is employed in Qiskit where the classical message 0110 is sent (Ref. table 3.2). The corresponding circuit and simulation results for bit flip noise $\eta_B = 0.1, 0.5$ and 0.9 are shown below.

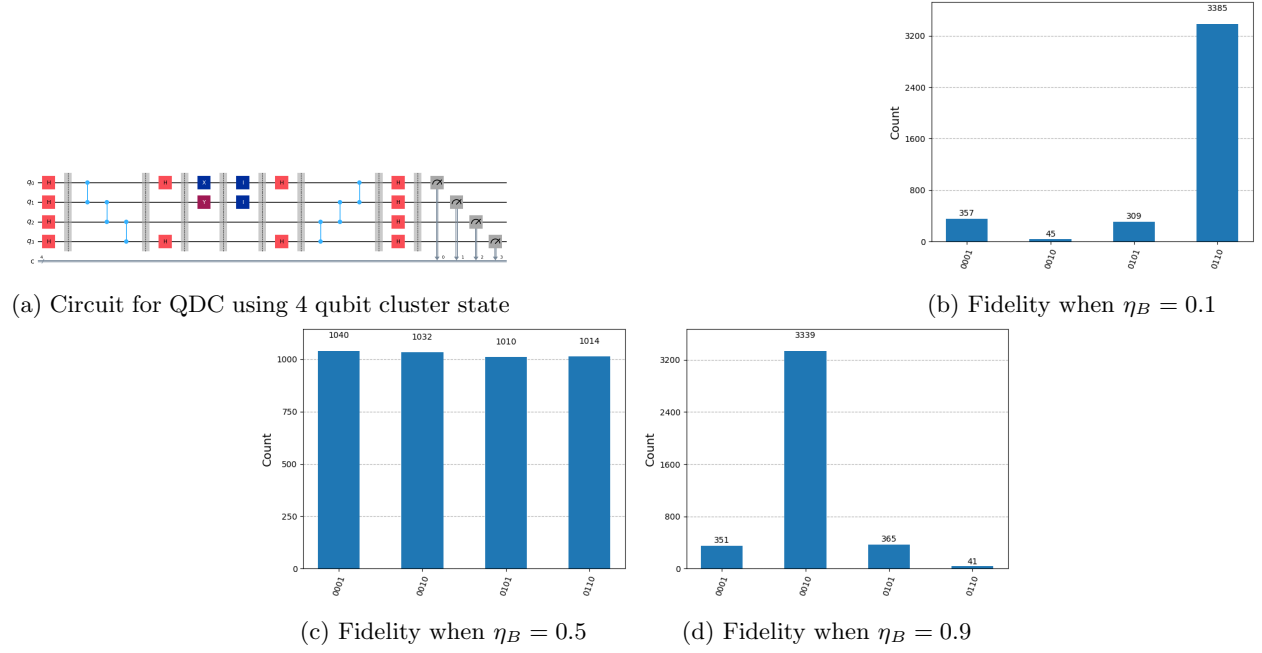


Figure 4.15: Implementation of QDC Protocol using 4 qubit cluster state when the message 0110 is sent through bit-phase flip channel

Analogously, the QDC protocol using 5-qubit cluster state under bit-phase flip noise is also employed in Qiskit where the classical message 01100 is sent (Ref. 3.3). The corresponding circuit and simulation results for bit-phase flip noise $\eta_B = 0.1, 0.5$ and 0.9 are shown below.

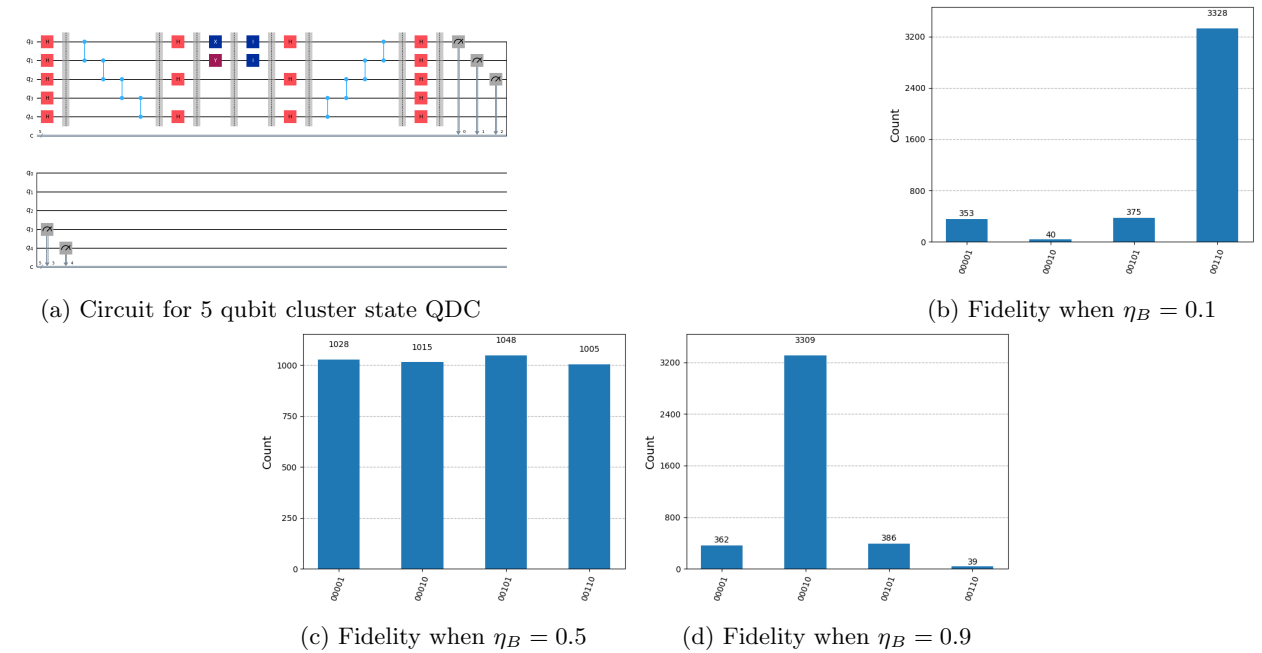


Figure 4.16: Implementation of QDC Protocol using 5 qubit cluster state when the message 01100 is sent through bit-phase flip channel

4.3.4 Implementation of QDC Protocol using 6 qubit cluster state

Finally, the dense coding protocol using six-qubit cluster-state is implemented under the bit-phase flip noise model. The transmitted qubits experience simultaneous bit and phase errors before being decoded by Bob. The resulting output fidelity is analysed over the entire range of the noise parameter and assessed the robustness of the different entangled quantum channels against combined Pauli errors.

The protocol simply follows the process as discussed in Sec (4.1.4) with a change in the implemented noise model. The Kraus operators corresponding to the bit-phase flip noise in this case are

$$K_{000} = (1 - \eta_F)^3 (I_1 \otimes I_4 \otimes I_6)$$

$$K_{001} = \eta_F (1 - \eta_F)^2 (I_1 \otimes I_4 \otimes Z_6)$$

$$K_{010} = \eta_F (1 - \eta_F)^2 (I_1 \otimes Y_4 \otimes I_6)$$

$$K_{011} = \eta_F^2 (1 - \eta_F) (I_1 \otimes Y_4 \otimes Y_6)$$

$$K_{100} = \eta_F (1 - \eta_F)^2 (Y_1 \otimes I_4 \otimes I_6)$$

$$K_{101} = \eta_F^2 (1 - \eta_F) (Y_1 \otimes I_4 \otimes Y_6)$$

$$K_{110} = \eta_F^2 (1 - \eta_F) (Y_1 \otimes Y_4 \otimes I_6)$$

$$K_{111} = \eta_F^3 (Y_1 \otimes Y_4 \otimes Y_6)$$

Using this operators the protocol is developed and implemented using QASM simulator, again and for $\eta_F = 0.1, 0.5$ and 0.9 the output fidelity is shown below. Here classical message 011001 is sent (Ref. Table3.4).

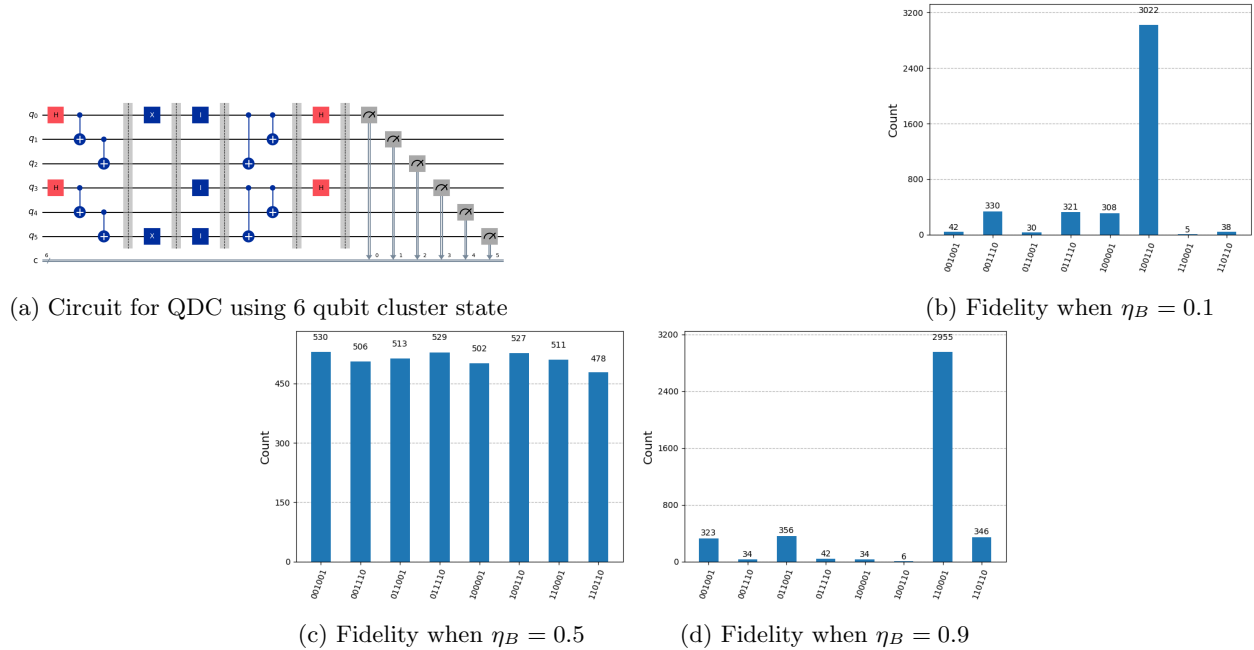


Figure 4.17: Implementation of QDC Protocol using 6 qubit cluster state when the message 011001 is sent through bit-phase flip channel

To check the robustness of the QDC protocol using various entangled quantum channels against bit-phase flip noise, the output state fidelity is evaluated as a function of the bit-phase-flip probability η_F . The following plot illustrates the variation of the output state fidelity for different entangled state dense coding protocols with respect to the noise parameter under application of bit-phase flip noise models.

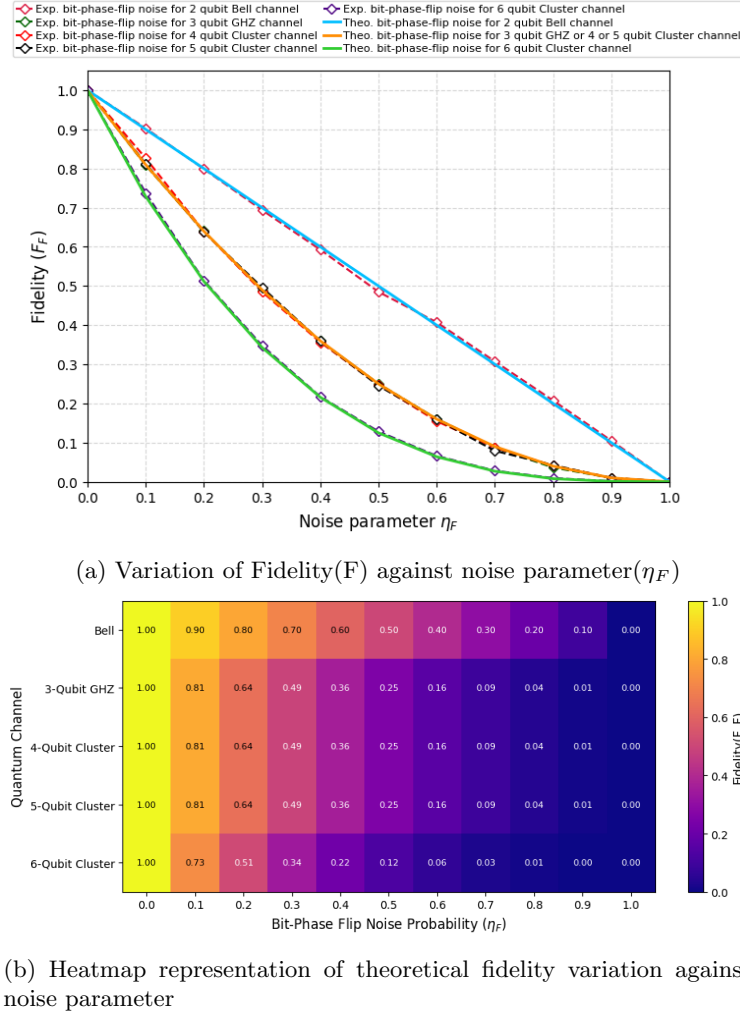
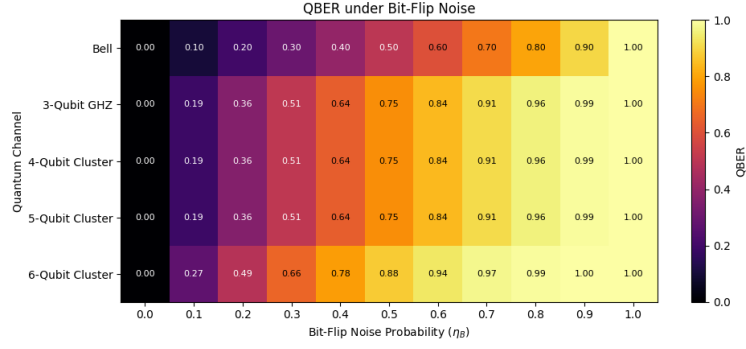


Figure 4.18: Comparison of robustness of different entangled channel QDC protocols under bit-phase flip noise

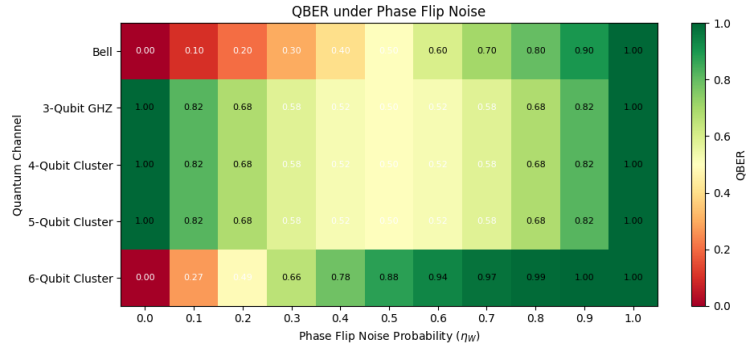
Remarks. For other encoding operators (and corresponding classical messages) as listed in the section (3.1.3) and tables (3.1, 3.2, 3.3, 3.4) the fate of fidelity will likely be the same as discussed above. Therefore *discussions regarding rest of the classical messages are generally avoided* while stating the protocol.

4.4 Analysis of different performance metrics affected by noise

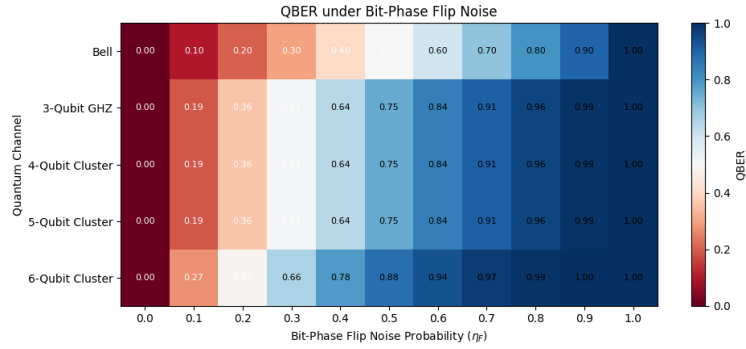
The performance metrics affected mostly by the noise are quantum state fidelity (Ref. Sec4) and QBER2.4.8. Since the output state fidelity has already been discussed in the previous sections, only QBER is to be discussed in this section. Lastly, QBER, (Ref. 2.4.8) is directly related to state fidelity for every noisy case study. Therefore QBER for different channels under different noise model using the respective fidelity data is summarized and represented using heatmaps below for a quick overview.



(a) when bit flip channel is considered



(b) when phase flip channel is considered



(c) when bit-phase flip channel is considered

Figure 4.19: Heatmap of QBER values for different dense coding protocols under different noisy channels

Chapter 5

Results & Discussions

Dense coding has long been recognized as one of the key protocols in quantum communication, enabling the transmission of classical information at rates beyond the classical limits by leveraging quantum resources. Motivated by this, we employ different entangled quantum states as the resource for QDC protocols and investigate their behavior under various noisy environments.

In the first half of the report, the fundamental concepts required for understanding quantum dense coding have been presented. These include brief discussion on quantum entanglement, performance metrics, Pauli-error-based noisy quantum channels, and dense coding protocols employing different entangled quantum states. The theoretical framework is supported by appropriate discussions and observations highlighting how various performance metrics change under different communication scenarios. The dense coding protocols were first analyzed in an ideal environment to establish their theoretical performance limits. Among the protocols investigated, the six-qubit cluster-state protocol produced the largest set of mutually orthogonal encoded states, enabling the transmission of six classical bits in a single communication round. While the five-qubit protocol is more efficient in terms of resource utilization, the six-qubit cluster state offers a larger communication capacity together with a more complex multipartite entanglement structure, making it a promising protocol for scalable quantum communication networks. It is also observed that increasing the number of entangled qubits increases the maximum amount of classical information that can be transmitted. Simultaneously, protocols with higher information capacity also exhibit greater implementation complexity in terms of used quantum gates. The analytical derivations developed throughout this work were found to be in exact agreement with the simulation results. Overall, these results confirm that maximally entangled quantum states provide a reliable foundation for deterministic quantum dense coding.

In the second half of this report, the quantum dense coding protocols are further investigated under various noisy quantum environments. Based on the state fidelity analysis, the Bell-state QDC protocol exhibits the highest information retention capability under bit-flip and bit-phase-flip noise models. In contrast, the three-qubit GHZ-state and the four- and five-qubit cluster-state protocols demonstrate superior communication capacity under the phase-flip noise model. It is also observed that an increase in circuit depth makes the protocols more susceptible to hardware-induced noise in practical quantum devices. Among the considered protocols, the Bell-state QDC circuit has the shallowest depth, whereas the cluster-state QDC circuits generally possess greater circuit depth. Furthermore, the quantum cost analysis reveals a direct relationship with the number of qubits, and the gate complexity. Finally, the heatmaps and state fidelity plots indicate that the effects of the bit-flip and bit-phase-flip noise models exhibit similar behavior for all the protocols considered in this work. For the source-codes of the results shown in this report refer [30].

In a nutshell, the findings reveal the strengths and limitations of different entangled channels under both ideal and realistic noisy environments, providing significant insights into the practical implementation of these protocols and the challenges encountered in achieving robust and efficient long-distance quantum information transmission.

Conclusion & Future Scopes

In conclusion, this work provides a systematic comparative analysis of quantum dense coding protocols under both ideal and noisy quantum environments, highlighting the trade-offs between communication capacity, resource requirements, and robustness against noise. Future work may focus on investigating additional noise models, such as amplitude damping, phase damping, and depolarizing channels. Exploring dense coding using other multipartite entangled states, such as the Brown et al. state, Borrás et al. state, graph states, Dicke states, and hypergraph states, would also be of significant practical interest. Moreover, reducing the gate complexity while achieving same deterministic information transmission can be a great area of study, as it could minimize the impact of noise infiltration. One can also incorporate quantum error correction and error mitigation techniques to resolve the issue of abrupt, incorrect information transmission. Extending these protocols to large-scale quantum communication networks has attracted considerable attention in recent years. Finally, extending the protocols to higher-dimensional systems and to evaluate their practical feasibility can be one of the major future challenge.

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